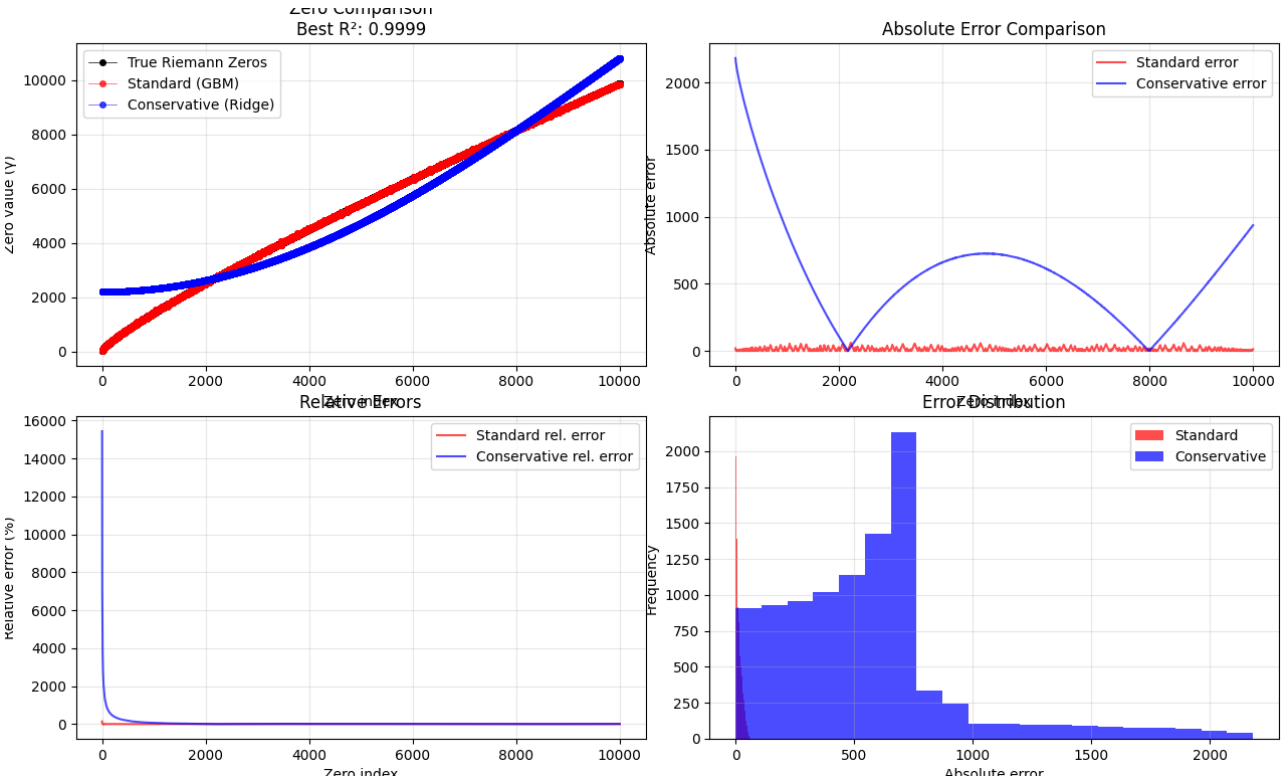
```
================================================================
==========================
QUANTUM OPERATOR → RIEMANN ZEROS
================================================================

Configuration:
  Prime limit: 150000
  Target zeros: 10000

----------------------------------------------------------------
--------
STEP 1: Generating true Riemann zeros (for validation)
----------------------------------------------------------------
--------
Generating first 10000 Riemann zeros using mpmath.zetazero()...
STEP 2: Building Hamiltonian
----------------------------------------------------------------
--------
Generating primes...
  Found 13848 primes up to 150000
Building numerical grid...
  Grid created with 20000 points
Building Hamiltonian matrix...

----------------------------------------------------------------
--------
STEP 3: Computing eigenvalues (this may take a moment)
----------------------------------------------------------------
--------

✓ Computed 10000 eigenvalues
  Range: [20.27, 5629255.15]

----------------------------------------------------------------
--------
STEP 4: Fitting models
----------------------------------------------------------------
--------

1. STANDARD ANALYSIS (GradientBoosting):
Analyzing 10000 examples

Testing Linear...
  CV R²: -3.6281 (±4.0496)
  Test R²: 0.9366

Testing GBM_Simple...
  CV R²: -1.5153 (±1.9356)
  Test R²: 0.9996

Testing GBM_Medium...
  CV R²: -1.1795 (±1.4267)
  Test R²: 0.9999

BEST MODEL: GBM_Medium (Test R² = 0.9999)
```

```
Stability check (bootstrap)...
  Bootstrap R²: 0.9379 (±0.0014)

OVERFITTING DIAGNOSTICS:
  Train R²: 1.0000
  Test R²: 0.9999
  Gap: 0.0000
  ✅ GOOD GENERALIZATION

2. CONSERVATIVE ANALYSIS (Ridge regression):

--------------------------------------------------------------------------------
STEP 5: Diagnostics
--------------------------------------------------------------------------------

EXTENDED DIAGNOSTICS
==================================================
Eigenvalue-zero correlation: 0.9686
Mean residual: 0.000000
Std of residuals: 683.457121
R² with quadratic terms: 0.9803

DIAGNOSTIC INDICATORS:
  High correlation: True (0.9686)
  Stable residuals: False (683.4571)
  Nonlinearity: False (0.9803)

FINAL ASSESSMENT:
  Standard Test R²: 0.9999
  Conservative Test R²: 0.9366
  Reliability: MEDIUM
```



STEP 6: Detailed Comparison

STANDARD MODEL:

#	Theory	Computed	Diff	Rel.Error(%)
1	14.134725	34.048436	19.913711	140.89
2	21.022040	35.156035	14.133995	67.23
3	25.010858	37.238676	12.227818	48.89
4	30.424876	40.538941	10.114065	33.24
5	32.935062	41.778188	8.843126	26.85
6	37.586178	45.954319	8.368141	22.26
7	40.918719	48.669902	7.751183	18.94
8	43.327073	49.802573	6.475500	14.95
9	48.005151	55.708267	7.703116	16.05
10	49.773832	55.708267	5.934434	11.92
20	77.144840	81.673682	4.528842	5.87
30	101.317851	105.378793	4.060942	4.01
40	122.946829	125.929175	2.982346	2.43
50	143.111846	141.297794	1.814052	1.27
60	163.030710	165.421309	2.390599	1.47
70	182.207078	183.547487	1.340408	0.74
80	201.264752	199.082352	2.182400	1.08
90	219.067596	216.523067	2.544529	1.16
100	236.524230	241.639252	5.115023	2.16
110	253.069987	244.350301	8.719686	3.45
120	269.970449	276.319159	6.348710	2.35
130	286.667445	277.112156	9.555290	3.33
140	302.696750	301.944048	0.752701	0.25
150	318.853104	312.574704	6.278400	1.97
160	334.211355	339.655291	5.443936	1.63
170	350.408419	339.655291	10.753128	3.07
180	366.212710	369.355422	3.142712	0.86
190	381.484469	385.390301	3.905832	1.02
200	396.381854	385.390301	10.991554	2.77
210	411.972268	418.491985	6.519717	1.58
220	427.208825	425.923284	1.285541	0.30
230	441.683199	425.923284	15.759915	3.57
240	456.328427	463.823978	7.495551	1.64
250	470.773655	463.823978	6.949677	1.48
260	485.539148	492.040563	6.501414	1.34
270	500.309085	506.847168	6.538083	1.31
280	513.668986	513.258227	0.410759	0.08
290	528.406214	513.258227	15.147987	2.87
300	541.847437	559.566376	17.718939	3.27
310	556.899476	559.566376	2.666900	0.48
320	570.051115	559.566376	10.484738	1.84
330	584.561706	609.568538	25.006832	4.28
340	598.493077	609.568538	11.075461	1.85

350	611.774210	609.568538	2.205672	0.36
360	626.019283	636.913692	10.894409	1.74
370	638.927938	639.968953	1.041015	0.16
380	653.649572	642.018229	11.631343	1.78
390	666.515148	642.018229	24.496919	3.68
400	679.742198	696.069809	16.327611	2.40
410	693.176970	697.685441	4.508471	0.65
420	706.184655	705.351169	0.833486	0.12
430	719.697101	705.351169	14.345932	1.99
440	732.818053	744.200770	11.382717	1.55
450	746.499306	744.200770	2.298536	0.31
460	758.900238	746.159170	12.741068	1.68
470	772.961618	746.159170	26.802447	3.47
480	785.739090	808.453757	22.714668	2.89
490	798.707570	808.453757	9.746187	1.22
500	811.184359	810.845710	0.338649	0.04
510	824.526294	810.845710	13.680584	1.66
520	837.347335	839.649273	2.301938	0.27
530	849.862274	851.171579	1.309304	0.15
540	863.189720	853.862622	9.327098	1.08
550	875.985285	855.835494	20.149791	2.30
560	888.475567	855.835494	32.640072	3.67
570	900.849740	928.350731	27.500991	3.05
580	912.823999	928.350731	15.526732	1.70
590	926.551553	931.071385	4.519832	0.49
600	939.024301	936.340498	2.683803	0.29
610	951.033249	936.340498	14.692750	1.54
620	963.567040	988.920252	25.353212	2.63
630	976.178502	988.920252	12.741750	1.31
640	988.992622	988.920252	0.072370	0.01
650	1001.349483	988.920252	12.429230	1.24
660	1013.058638	988.920252	24.138386	2.38
670	1025.707944	988.920252	36.787692	3.59
680	1038.087752	1079.824222	41.736470	4.02
690	1049.996284	1079.824222	29.827938	2.84
700	1062.915382	1079.824222	16.908841	1.59
710	1074.747771	1079.824222	5.076451	0.47
720	1086.911999	1082.304072	4.607927	0.42
730	1099.360667	1100.469193	1.108526	0.10
740	1111.443505	1104.576186	6.867319	0.62
750	1123.101117	1121.980209	1.120908	0.10
760	1135.562214	1136.393453	0.831240	0.07
770	1147.501777	1138.728526	8.773250	0.76
780	1159.480657	1138.728526	20.752131	1.79
790	1172.120682	1202.520851	30.400169	2.59
800	1183.712775	1202.520851	18.808075	1.59
810	1196.034672	1204.478332	8.443660	0.71
820	1208.471460	1210.527063	2.055603	0.17
830	1219.614471	1217.158644	2.455827	0.20
840	1231.562274	1217.158644	14.403630	1.17
850	1243.538147	1220.813262	22.724884	1.83
860	1255.408231	1223.306750	32.101480	2.56
870	1267.200346	1223.306750	43.893595	3.46
880	1279.332843	1316.016090	36.683247	2.87
890	1290.417708	1316.016090	25.598382	1.98
900	1302.346742	1318.725953	16.379211	1.26

910	1314.052566	1322.000490	7.947924	0.60
920	1325.981970	1324.942933	1.039037	0.08
930	1337.688792	1329.488442	8.200350	0.61
940	1349.085194	1329.488442	19.596752	1.45
950	1361.393075	1329.488442	31.904633	2.34
960	1373.202915	1374.671111	1.468196	0.11
970	1384.444416	1374.671111	9.773305	0.71
980	1396.544163	1412.861311	16.317147	1.17
990	1408.136307	1412.861311	4.725003	0.34
1000	1419.422481	1412.861311	6.561170	0.46
1010	1431.218314	1416.806170	14.412144	1.01
1020	1442.911060	1423.276537	19.634523	1.36
1030	1454.226011	1423.276537	30.949474	2.13
1040	1465.889558	1423.276537	42.613021	2.91
1050	1477.435191	1526.089610	48.654419	3.29
1060	1489.252467	1526.089610	36.837143	2.47
1070	1500.351952	1526.089610	25.737658	1.72
1080	1511.923005	1532.706131	20.783125	1.37
1090	1523.486047	1536.536132	13.050085	0.86
1100	1535.042277	1536.536132	1.493854	0.10
1110	1546.091198	1542.724605	3.366593	0.22
1120	1557.394507	1556.672606	0.721901	0.05
1130	1569.022478	1561.402514	7.619964	0.49
1140	1580.372082	1563.923329	16.448753	1.04
1150	1591.293884	1563.923329	27.370556	1.72
1160	1602.815405	1563.923329	38.892076	2.43
1170	1614.710628	1642.463463	27.752835	1.72
1180	1625.987020	1642.463463	16.476443	1.01
1190	1637.089190	1645.319317	8.230127	0.50
1200	1648.270598	1650.185454	1.914855	0.12
1210	1660.019980	1662.493981	2.474001	0.15
1220	1671.047182	1662.493981	8.553201	0.51
1230	1681.812320	1662.493981	19.318339	1.15
1240	1693.053021	1666.138667	26.914354	1.59
1250	1704.291412	1666.138667	38.152746	2.24
1260	1715.825283	1666.138667	49.686616	2.90
1270	1726.605701	1772.595424	45.989723	2.66
1280	1737.893885	1772.595424	34.701539	2.00
1290	1748.563987	1772.595424	24.031437	1.37
1300	1760.215166	1776.540583	16.325417	0.93
1310	1771.212945	1776.540583	5.327638	0.30
1320	1782.439067	1784.616581	2.177514	0.12
1330	1793.525391	1786.497463	7.027928	0.39
1340	1804.904161	1786.497463	18.406698	1.02
1350	1816.338611	1786.497463	29.841148	1.64
1360	1826.795455	1786.497463	40.297992	2.21
1370	1838.086943	1885.958357	47.871414	2.60
1380	1848.722761	1885.958357	37.235596	2.01
1390	1860.541388	1885.958357	25.416969	1.37
1400	1871.416043	1885.958357	14.542314	0.78
1410	1882.321975	1885.958357	3.636382	0.19
1420	1892.867701	1885.958357	6.909344	0.37
1430	1904.089325	1885.958357	18.130968	0.95
1440	1915.054629	1925.983506	10.928877	0.57
1450	1926.211583	1925.983506	0.228077	0.01
1460	1936.873094	1927.838417	9.034677	0.47

1470	1947.917048	1941.856211	6.060837	0.31
1480	1958.851883	1959.125135	0.273252	0.01
1490	1970.305010	1971.475253	1.170243	0.06
1500	1980.910043	1993.949889	13.039846	0.66
1510	1992.001706	1993.949889	1.948182	0.10
1520	2002.896408	1993.949889	8.946520	0.45
1530	2013.872098	2020.344491	6.472393	0.32
1540	2024.488262	2020.344491	4.143772	0.20
1550	2035.077440	2040.712405	5.634965	0.28
1560	2045.802760	2042.363078	3.439682	0.17
1570	2056.620867	2042.363078	14.257789	0.69
1580	2068.345639	2086.436734	18.091096	0.87
1590	2078.939288	2086.436734	7.497446	0.36
1600	2090.036207	2094.101845	4.065638	0.19
1610	2100.593745	2094.101845	6.491900	0.31
1620	2110.822372	2094.101845	16.720527	0.79
1630	2122.465151	2094.101845	28.363306	1.34
1640	2132.798516	2094.101845	38.696671	1.81
1650	2143.374444	2147.036342	3.661898	0.17
1660	2154.263624	2147.036342	7.227282	0.34
1670	2165.011437	2147.036342	17.975095	0.83
1680	2176.204119	2188.724970	12.520850	0.58
1690	2187.085691	2188.724970	1.639279	0.07
1700	2197.258747	2188.724970	8.533777	0.39
1710	2207.802741	2188.724970	19.077772	0.86
1720	2218.521007	2256.525118	38.004111	1.71
1730	2229.127959	2256.525118	27.397159	1.23
1740	2240.021209	2256.525118	16.503908	0.74
1750	2251.046642	2256.525118	5.478476	0.24
1760	2261.660064	2256.525118	5.134947	0.23
1770	2272.049016	2256.525118	15.523898	0.68
1780	2282.690375	2256.525118	26.165257	1.15
1790	2293.896145	2330.625637	36.729492	1.60
1800	2304.305139	2330.625637	26.320498	1.14
1810	2314.621182	2330.625637	16.004455	0.69
1820	2325.557096	2330.625637	5.068540	0.22
1830	2335.867154	2334.578574	1.288580	0.06
1840	2346.464120	2342.637070	3.827049	0.16
1850	2357.608316	2355.571549	2.036767	0.09
1860	2367.957702	2355.571549	12.386153	0.52
1870	2378.212962	2358.598113	19.614848	0.82
1880	2388.748208	2358.598113	30.150094	1.26
1890	2399.433833	2358.598113	40.835720	1.70
1900	2409.881603	2442.918261	33.036659	1.37
1910	2421.142944	2442.918261	21.775317	0.90
1920	2431.611135	2445.541334	13.930199	0.57
1930	2441.929837	2445.541334	3.611498	0.15
1940	2452.283366	2455.936767	3.653401	0.15
1950	2462.464295	2457.084297	5.379998	0.22
1960	2473.107125	2481.086953	7.979828	0.32
1970	2484.503238	2481.086953	3.416286	0.14
1980	2494.866178	2481.086953	13.779225	0.55
1990	2504.772153	2481.086953	23.685200	0.95
2000	2515.286483	2533.817587	18.531104	0.74
2010	2525.687993	2533.817587	8.129594	0.32
2020	2536.091885	2533.817587	2.274298	0.09

2030	2546.761141	2533.817587	12.943554	0.51
2040	2558.009106	2537.193802	20.815304	0.81
2050	2567.958600	2537.193802	30.764798	1.20
2060	2578.070130	2537.193802	40.876328	1.59
2070	2588.368210	2640.957443	52.589233	2.03
2080	2598.598768	2640.957443	42.358675	1.63
2090	2609.552087	2640.957443	31.405357	1.20
2100	2620.279394	2640.957443	20.678050	0.79
2110	2629.994801	2644.793691	14.798890	0.56
2120	2640.462943	2644.793691	4.330748	0.16
2130	2651.227331	2644.793691	6.433641	0.24
2140	2661.582574	2672.566577	10.984003	0.41
2150	2671.918383	2672.566577	0.648195	0.02
2160	2681.841051	2686.653520	4.812468	0.18
2170	2692.368719	2692.059965	0.308753	0.01
2180	2703.143658	2692.059965	11.083692	0.41
2190	2713.525289	2692.059965	21.465324	0.79
2200	2723.768929	2692.059965	31.708963	1.16
2210	2734.200783	2692.059965	42.140817	1.54
2220	2743.953470	2692.059965	51.893504	1.89
2230	2754.458663	2692.059965	62.398698	2.27
2240	2765.407691	2817.425173	52.017482	1.88
2250	2775.725707	2817.425173	41.699466	1.50
2260	2786.043336	2817.425173	31.381838	1.13
2270	2796.034762	2817.425173	21.390412	0.77
2280	2806.002732	2817.425173	11.422441	0.41
2290	2816.337955	2817.425173	1.087218	0.04
2300	2826.712170	2817.425173	9.286997	0.33
2310	2837.014472	2817.425173	19.589299	0.69
2320	2847.535936	2855.601244	8.065309	0.28
2330	2857.736140	2855.601244	2.134895	0.07
2340	2868.224164	2855.601244	12.622919	0.44
2350	2878.527003	2855.601244	22.925758	0.80
2360	2888.394382	2917.333337	28.938955	1.00
2370	2898.541353	2917.333337	18.791984	0.65
2380	2908.658046	2917.333337	8.675292	0.30
2390	2919.321772	2917.333337	1.988435	0.07
2400	2930.004366	2929.245569	0.758797	0.03
2410	2939.847529	2929.245569	10.601960	0.36
2420	2949.790143	2938.738167	11.051976	0.37
2430	2960.033618	2938.738167	21.295451	0.72
2440	2970.049930	2938.738167	31.311764	1.05
2450	2980.487381	3022.277260	41.789879	1.40
2460	2990.423271	3022.277260	31.853989	1.07
2470	3001.023331	3022.277260	21.253929	0.71
2480	3011.344380	3027.716295	16.371915	0.54
2490	3021.420401	3030.500249	9.079848	0.30
2500	3031.289217	3030.500249	0.788968	0.03
2510	3041.256804	3044.469954	3.213151	0.11
2520	3051.342383	3053.308836	1.966452	0.06
2530	3061.837584	3053.308836	8.528749	0.28
2540	3072.000993	3053.308836	18.692158	0.61
2550	3082.163164	3055.804570	26.358594	0.86
2560	3092.687667	3055.804570	36.883097	1.19
2570	3102.436089	3134.631051	32.194962	1.04
2580	3112.840738	3134.631051	21.790314	0.70

2590	3122.610129	3134.631051	12.020923	0.38
2600	3132.698331	3134.631051	1.932720	0.06
2610	3142.979457	3134.631051	8.348405	0.27
2620	3152.994560	3136.984478	16.010082	0.51
2630	3163.311960	3136.984478	26.327482	0.83
2640	3173.011634	3205.251588	32.239954	1.02
2650	3183.205673	3205.251588	22.045915	0.69
2660	3193.557208	3207.266538	13.709329	0.43
2670	3203.630891	3207.266538	3.635647	0.11
2680	3213.606499	3211.899831	1.706668	0.05
2690	3223.372566	3216.779858	6.592708	0.20
2700	3233.913361	3216.779858	17.133502	0.53
2710	3243.501653	3246.083945	2.582292	0.08
2720	3253.369017	3246.083945	7.285072	0.22
2730	3263.784940	3278.070518	14.285577	0.44
2740	3274.059953	3278.070518	4.010565	0.12
2750	3284.499863	3278.070518	6.429345	0.20
2760	3293.975869	3289.370546	4.605324	0.14
2770	3303.813719	3303.322511	0.491208	0.01
2780	3313.842662	3310.467185	3.375477	0.10
2790	3323.875125	3310.467185	13.407940	0.40
2800	3333.781383	3350.697338	16.915955	0.51
2810	3343.604909	3350.697338	7.092430	0.21
2820	3354.278231	3350.697338	3.580893	0.11
2830	3364.355764	3350.697338	13.658425	0.41
2840	3374.427472	3350.697338	23.730134	0.70
2850	3383.871212	3400.166837	16.295624	0.48
2860	3394.137666	3400.166837	6.029170	0.18
2870	3404.362762	3404.354000	0.008762	0.00
2880	3413.710819	3404.979590	8.731229	0.26
2890	3423.611395	3425.747782	2.136387	0.06
2900	3433.769710	3425.747782	8.021929	0.23
2910	3444.036571	3425.747782	18.288789	0.53
2920	3453.677894	3425.747782	27.930112	0.81
2930	3464.083832	3425.747782	38.336051	1.11
2940	3473.814882	3501.989855	28.174973	0.81
2950	3484.159388	3501.989855	17.830467	0.51
2960	3493.720613	3501.989855	8.269242	0.24
2970	3503.636216	3501.989855	1.646361	0.05
2980	3513.877574	3518.949422	5.071848	0.14
2990	3523.635836	3518.949422	4.686414	0.13
3000	3533.328243	3521.921629	11.406615	0.32
3010	3543.322307	3521.921629	21.400678	0.60
3020	3553.840681	3521.921629	31.919052	0.90
3030	3563.215121	3609.615687	46.400566	1.30
3040	3573.079804	3609.615687	36.535883	1.02
3050	3582.923042	3609.615687	26.692645	0.74
3060	3593.028559	3609.615687	16.587129	0.46
3070	3602.418894	3612.825971	10.407077	0.29
3080	3612.324193	3612.825971	0.501779	0.01
3090	3622.179275	3612.825971	9.353303	0.26
3100	3632.478178	3612.825971	19.652207	0.54
3110	3641.707123	3612.825971	28.881152	0.79
3120	3651.783523	3673.935935	22.152412	0.61
3130	3662.368949	3673.935935	11.566986	0.32
3140	3672.349351	3673.935935	1.586584	0.04

3150	3682.418731	3680.506981	1.911750	0.05
3160	3691.756940	3688.195589	3.561350	0.10
3170	3702.050757	3705.863870	3.813113	0.10
3180	3711.404818	3709.708874	1.695944	0.05
3190	3721.296040	3709.708874	11.587166	0.31
3200	3731.075159	3709.708874	21.366285	0.57
3210	3740.875154	3712.564261	28.310893	0.76
3220	3750.303158	3712.564261	37.738898	1.01
3230	3760.582729	3712.564261	48.018469	1.28
3240	3770.105498	3817.723708	47.618210	1.26
3250	3780.226486	3817.723708	37.497222	0.99
3260	3790.041777	3817.723708	27.681931	0.73
3270	3800.132426	3817.723708	17.591282	0.46
3280	3809.874248	3821.540546	11.666298	0.31
3290	3819.414499	3821.540546	2.126048	0.06
3300	3829.144810	3822.381207	6.763603	0.18
3310	3839.426268	3822.381207	17.045061	0.44
3320	3848.415301	3859.896865	11.481564	0.30
3330	3858.448586	3859.896865	1.448279	0.04
3340	3868.412859	3870.158001	1.745142	0.05
3350	3878.256175	3877.396196	0.859979	0.02
3360	3887.352014	3892.608543	5.256529	0.14
3370	3897.870750	3899.509140	1.638390	0.04
3380	3908.188165	3899.509140	8.679025	0.22
3390	3917.658166	3926.777328	9.119162	0.23
3400	3927.080517	3926.777328	0.303189	0.01
3410	3936.974234	3926.777328	10.196906	0.26
3420	3946.878984	3931.061644	15.817340	0.40
3430	3956.420217	3931.061644	25.358573	0.64
3440	3966.138715	3931.061644	35.077071	0.88
3450	3976.066183	3931.061644	45.004539	1.13
3460	3985.098725	3931.061644	54.037081	1.36
3470	3995.168460	4044.909944	49.741485	1.25
3480	4004.806731	4044.909944	40.103213	1.00
3490	4014.657295	4044.909944	30.252649	0.75
3500	4023.905151	4044.909944	21.004793	0.52
3510	4034.520820	4048.589864	14.069043	0.35
3520	4044.322287	4048.589864	4.267577	0.11
3530	4053.566272	4048.589864	4.976408	0.12
3540	4063.157725	4048.589864	14.567862	0.36
3550	4073.126450	4099.679030	26.552580	0.65
3560	4082.588348	4099.679030	17.090682	0.42
3570	4092.297206	4099.679030	7.381824	0.18
3580	4102.037363	4101.906932	0.130430	0.00
3590	4111.859845	4105.451667	6.408179	0.16
3600	4121.348657	4130.214110	8.865453	0.22
3610	4131.136980	4130.214110	0.922870	0.02
3620	4140.892830	4141.282810	0.389980	0.01
3630	4150.445098	4141.282810	9.162288	0.22
3640	4159.921666	4141.282810	18.638856	0.45
3650	4170.482806	4141.282810	29.199996	0.70
3660	4179.574699	4202.053655	22.478955	0.54
3670	4189.615280	4202.053655	12.438375	0.30
3680	4199.335193	4202.053655	2.718462	0.06
3690	4208.734782	4202.053655	6.681128	0.16
3700	4218.230789	4243.066187	24.835398	0.59

3710	4227.984705	4243.066187	15.081482	0.36
3720	4237.343105	4243.066187	5.723082	0.14
3730	4247.050666	4247.335456	0.284789	0.01
3740	4256.903916	4247.335456	9.568461	0.22
3750	4266.175526	4247.335456	18.840071	0.44
3760	4275.863648	4247.335456	28.528192	0.67
3770	4285.629054	4247.335456	38.293598	0.89
3780	4295.348211	4342.252230	46.904019	1.09
3790	4304.617528	4342.252230	37.634702	0.87
3800	4314.799283	4342.252230	27.452947	0.64
3810	4324.546748	4342.252230	17.705482	0.41
3820	4334.020581	4342.252230	8.231649	0.19
3830	4343.563201	4342.252230	1.310971	0.03
3840	4353.700190	4349.278272	4.421917	0.10
3850	4362.761583	4361.059863	1.701720	0.04
3860	4372.227907	4372.844054	0.616147	0.01
3870	4382.125949	4381.269682	0.856267	0.02
3880	4391.021213	4383.761739	7.259474	0.17
3890	4401.175472	4383.761739	17.413732	0.40
3900	4410.491142	4383.761739	26.729403	0.61
3910	4419.922458	4383.761739	36.160718	0.82
3920	4429.488010	4464.048619	34.560609	0.78
3930	4439.557561	4464.048619	24.491058	0.55
3940	4448.301901	4464.048619	15.746718	0.35
3950	4458.188749	4464.048619	5.859869	0.13
3960	4467.739817	4464.048619	3.691198	0.08
3970	4477.015857	4464.048619	12.967238	0.29
3980	4487.084191	4464.048619	23.035573	0.51
3990	4496.893452	4529.478230	32.584778	0.72
4000	4506.311497	4529.478230	23.166733	0.51
4010	4516.089845	4529.478230	13.388384	0.30
4020	4525.790882	4529.478230	3.687347	0.08
4030	4534.935556	4532.252043	2.683513	0.06
4040	4544.720271	4536.420135	8.300137	0.18
4050	4554.197951	4536.420135	17.777816	0.39
4060	4563.416046	4564.825388	1.409342	0.03
4070	4573.246173	4590.251284	17.005111	0.37
4080	4582.568249	4590.251284	7.683035	0.17
4090	4592.049309	4590.455152	1.594157	0.03
4100	4601.602962	4597.972538	3.630424	0.08
4110	4611.484674	4610.961949	0.522725	0.01
4120	4620.544366	4624.805500	4.261135	0.09
4130	4629.888783	4624.805500	5.083283	0.11
4140	4639.715381	4641.180360	1.464979	0.03
4150	4648.564887	4641.180360	7.384527	0.16
4160	4658.394950	4662.879341	4.484391	0.10
4170	4668.055178	4668.022882	0.032296	0.00
4180	4677.930219	4668.022882	9.907337	0.21
4190	4687.072178	4668.022882	19.049296	0.41
4200	4697.354147	4713.953925	16.599777	0.35
4210	4706.468571	4713.953925	7.485354	0.16
4220	4715.952622	4718.110510	2.157888	0.05
4230	4725.487197	4718.110510	7.376687	0.16
4240	4734.527818	4718.110510	16.417308	0.35
4250	4744.421811	4764.868523	20.446711	0.43
4260	4753.918836	4764.868523	10.949687	0.23

4270	4763.158189	4764.868523	1.710334	0.04
4280	4772.644939	4764.868523	7.776416	0.16
4290	4782.356874	4764.868523	17.488352	0.37
4300	4791.291171	4764.868523	26.422648	0.55
4310	4800.671002	4822.259120	21.588118	0.45
4320	4810.531881	4822.259120	11.727238	0.24
4330	4819.754628	4822.259120	2.504492	0.05
4340	4829.386310	4822.259120	7.127190	0.15
4350	4838.660484	4822.259120	16.401364	0.34
4360	4848.107405	4822.259120	25.848286	0.53
4370	4857.815380	4890.341729	32.526349	0.67
4380	4866.954021	4890.341729	23.387708	0.48
4390	4876.743142	4890.341729	13.598587	0.28
4400	4886.270479	4890.341729	4.071250	0.08
4410	4895.963740	4893.278965	2.684775	0.05
4420	4905.142489	4907.578033	2.435544	0.05
4430	4914.792646	4907.578033	7.214613	0.15
4440	4924.317098	4907.578033	16.739065	0.34
4450	4933.068126	4910.032197	23.035929	0.47
4460	4942.976896	4910.032197	32.944700	0.67
4470	4951.878798	4988.979590	37.100792	0.75
4480	4961.688463	4988.979590	27.291127	0.55
4490	4971.339246	4988.979590	17.640345	0.35
4500	4980.118763	4991.931242	11.812479	0.24
4510	4990.345480	4991.931242	1.585761	0.03
4520	4999.329681	4993.346408	5.983273	0.12
4530	5008.626200	4993.346408	15.279791	0.31
4540	5018.084914	5024.679057	6.594143	0.13
4550	5027.221564	5026.795796	0.425767	0.01
4560	5036.841667	5035.859922	0.981745	0.02
4570	5045.941676	5035.859922	10.081754	0.20
4580	5055.192672	5035.859922	19.332751	0.38
4590	5064.628973	5038.925865	25.703108	0.51
4600	5074.356062	5038.925865	35.430198	0.70
4610	5083.769849	5038.925865	44.843984	0.88
4620	5092.837760	5138.665496	45.827736	0.90
4630	5102.645949	5138.665496	36.019548	0.71
4640	5111.503125	5138.665496	27.162372	0.53
4650	5121.083490	5138.665496	17.582006	0.34
4660	5131.043899	5142.322136	11.278237	0.22
4670	5139.679126	5142.322136	2.643010	0.05
4680	5149.768260	5142.322136	7.446125	0.14
4690	5158.954824	5142.322136	16.632688	0.32
4700	5167.990828	5142.322136	25.668692	0.50
4710	5177.609630	5142.322136	35.287494	0.68
4720	5186.709473	5142.322136	44.387337	0.86
4730	5196.707324	5142.322136	54.385189	1.05
4740	5205.900596	5259.248911	53.348315	1.02
4750	5214.775253	5259.248911	44.473658	0.85
4760	5224.457463	5259.248911	34.791447	0.67
4770	5233.658115	5259.248911	25.590796	0.49
4780	5242.792985	5259.248911	16.455925	0.31
4790	5252.522038	5259.248911	6.726873	0.13
4800	5261.180476	5259.248911	1.931565	0.04
4810	5270.886228	5263.459943	7.426284	0.14
4820	5280.315446	5274.383232	5.932214	0.11

4830	5289.353245	5274.383232	14.970014	0.28
4840	5299.274679	5312.069201	12.794522	0.24
4850	5308.072790	5312.069201	3.996410	0.08
4860	5317.139735	5320.001716	2.861981	0.05
4870	5326.867764	5320.001716	6.866047	0.13
4880	5335.947926	5320.001716	15.946210	0.30
4890	5345.528297	5320.001716	25.526581	0.48
4900	5354.773480	5320.001716	34.771763	0.65
4910	5363.717280	5320.001716	43.715564	0.82
4920	5373.219973	5414.287630	41.067657	0.76
4930	5382.809993	5414.287630	31.477637	0.58
4940	5391.505966	5414.287630	22.781665	0.42
4950	5401.194496	5414.287630	13.093134	0.24
4960	5410.101872	5414.287630	4.185758	0.08
4970	5419.647914	5414.287630	5.360284	0.10
4980	5429.307069	5436.545412	7.238343	0.13
4990	5438.486078	5436.545412	1.940665	0.04
5000	5447.861998	5436.545412	11.316586	0.21
5010	5457.254336	5436.545412	20.708923	0.38
5020	5466.418306	5436.545412	29.872894	0.55
5030	5475.749437	5511.983427	36.233990	0.66
5040	5485.379530	5511.983427	26.603897	0.48
5050	5494.038481	5511.983427	17.944946	0.33
5060	5503.522536	5511.983427	8.460891	0.15
5070	5513.109958	5511.983427	1.126531	0.02
5080	5522.249631	5516.224553	6.025078	0.11
5090	5531.929613	5516.224553	15.705060	0.28
5100	5540.694583	5516.224553	24.470030	0.44
5110	5549.834823	5577.277014	27.442191	0.49
5120	5559.248179	5577.277014	18.028835	0.32
5130	5568.798580	5577.277014	8.478434	0.15
5140	5577.672277	5577.969871	0.297594	0.01
5150	5587.160081	5577.969871	9.190210	0.16
5160	5596.227737	5577.969871	18.257866	0.33
5170	5605.644654	5628.360396	22.715742	0.41
5180	5614.492288	5628.360396	13.868108	0.25
5190	5623.777120	5628.360396	4.583276	0.08
5200	5632.847100	5636.983620	4.136520	0.07
5210	5642.463399	5636.983620	5.479779	0.10
5220	5651.883418	5636.983620	14.899799	0.26
5230	5660.630423	5674.617556	13.987134	0.25
5240	5669.978564	5674.617556	4.638993	0.08
5250	5679.113904	5674.617556	4.496348	0.08
5260	5688.363844	5674.617556	13.746288	0.24
5270	5697.902080	5702.061568	4.159489	0.07
5280	5706.596076	5702.061568	4.534508	0.08
5290	5716.333560	5726.933344	10.599783	0.19
5300	5725.671141	5726.933344	1.262203	0.02
5310	5734.289093	5727.509130	6.779962	0.12
5320	5743.891994	5748.153187	4.261193	0.07
5330	5752.725353	5757.188867	4.463514	0.08
5340	5762.451907	5765.838082	3.386175	0.06
5350	5771.717743	5771.322260	0.395483	0.01
5360	5780.378597	5783.065169	2.686572	0.05
5370	5789.954351	5786.266012	3.688339	0.06
5380	5798.939664	5786.266012	12.673652	0.22

5390	5807.786383	5786.266012	21.520371	0.37
5400	5817.687142	5831.577273	13.890132	0.24
5410	5826.437166	5831.577273	5.140107	0.09
5420	5835.952598	5844.781675	8.829077	0.15
5430	5845.043541	5845.937362	0.893821	0.02
5440	5854.191540	5846.010681	8.180859	0.14
5450	5863.094039	5846.010681	17.083358	0.29
5460	5872.402795	5846.010681	26.392114	0.45
5470	5881.825024	5846.010681	35.814343	0.61
5480	5890.842689	5915.639112	24.796422	0.42
5490	5900.026131	5915.639112	15.612981	0.26
5500	5909.432316	5915.639112	6.206796	0.11
5510	5917.975418	5915.639112	2.336306	0.04
5520	5927.479356	5932.402250	4.922893	0.08
5530	5936.465147	5932.402250	4.062898	0.07
5540	5946.143332	5932.402250	13.741082	0.23
5550	5955.342138	5932.402250	22.939889	0.39
5560	5964.146533	5932.402250	31.744284	0.53
5570	5973.620902	6016.180025	42.559122	0.71
5580	5982.914977	6016.180025	33.265048	0.56
5590	5991.472985	6016.180025	24.707039	0.41
5600	6001.495519	6016.180025	14.684505	0.24
5610	6010.190820	6016.180025	5.989204	0.10
5620	6019.650286	6018.846018	0.804268	0.01
5630	6029.083879	6025.545364	3.538515	0.06
5640	6037.621815	6025.545364	12.076451	0.20
5650	6046.796932	6049.785352	2.988420	0.05
5660	6056.260886	6049.785352	6.475534	0.11
5670	6065.104447	6049.785352	15.319094	0.25
5680	6074.835752	6049.785352	25.050400	0.41
5690	6083.644017	6112.006857	28.362841	0.47
5700	6093.192335	6112.006857	18.814522	0.31
5710	6101.542636	6112.006857	10.464221	0.17
5720	6110.830996	6112.006857	1.175861	0.02
5730	6120.281535	6117.004424	3.277110	0.05
5740	6129.268752	6117.004424	12.264327	0.20
5750	6138.445601	6120.440864	18.004738	0.29
5760	6147.070023	6120.440864	26.629160	0.43
5770	6156.737464	6120.440864	36.296601	0.59
5780	6166.200513	6120.440864	45.759650	0.74
5790	6174.747542	6216.727607	41.980064	0.68
5800	6184.336317	6216.727607	32.391290	0.52
5810	6193.133431	6216.727607	23.594175	0.38
5820	6201.832570	6219.866983	18.034414	0.29
5830	6211.207817	6219.866983	8.659167	0.14
5840	6220.276866	6219.866983	0.409883	0.01
5850	6229.761584	6229.370330	0.391254	0.01
5860	6238.507790	6243.176293	4.668503	0.07
5870	6247.765900	6243.886879	3.879021	0.06
5880	6256.737988	6243.886879	12.851109	0.21
5890	6265.972125	6243.886879	22.085246	0.35
5900	6275.133982	6290.373983	15.240001	0.24
5910	6284.054280	6290.373983	6.319703	0.10
5920	6293.297336	6291.558264	1.739072	0.03
5930	6302.230260	6309.499500	7.269240	0.12
5940	6311.201066	6314.986424	3.785359	0.06

5950	6320.918880	6324.896521	3.977641	0.06
5960	6329.665095	6324.896521	4.768574	0.08
5970	6339.041957	6324.896521	14.145436	0.22
5980	6347.512775	6331.867778	15.644997	0.25
5990	6356.984107	6331.867778	25.116329	0.40
6000	6365.852321	6331.867778	33.984543	0.53
6010	6375.032405	6331.867778	43.164627	0.68
6020	6384.459219	6434.545537	50.086317	0.78
6030	6392.718575	6434.545537	41.826961	0.65
6040	6402.175291	6434.545537	32.370246	0.51
6050	6411.384469	6434.545537	23.161067	0.36
6060	6420.182579	6434.545537	14.362958	0.22
6070	6429.468849	6442.296356	12.827507	0.20
6080	6438.740481	6442.296356	3.555875	0.06
6090	6447.474618	6442.991821	4.482797	0.07
6100	6456.243758	6457.388020	1.144261	0.02
6110	6465.743874	6457.388020	8.355854	0.13
6120	6475.021493	6457.388020	17.633474	0.27
6130	6483.450190	6457.388020	26.062170	0.40
6140	6492.813487	6501.780119	8.966632	0.14
6150	6501.665184	6501.780119	0.114934	0.00
6160	6510.892054	6501.780119	9.111935	0.14
6170	6520.320493	6538.802099	18.481606	0.28
6180	6529.103610	6538.802099	9.698489	0.15
6190	6538.142600	6538.802099	0.659499	0.01
6200	6546.926187	6545.957715	0.968472	0.01
6210	6556.290164	6545.957715	10.332449	0.16
6220	6565.225897	6571.484774	6.258877	0.10
6230	6574.230085	6574.376720	0.146635	0.00
6240	6583.647990	6574.376720	9.271270	0.14
6250	6592.144011	6574.376720	17.767292	0.27
6260	6601.347458	6574.376720	26.970738	0.41
6270	6610.094416	6627.907446	17.813030	0.27
6280	6619.598204	6627.907446	8.309242	0.13
6290	6628.061837	6627.907446	0.154391	0.00
6300	6637.625364	6627.907446	9.717918	0.15
6310	6646.611796	6630.728011	15.883784	0.24
6320	6655.334573	6630.728011	24.606562	0.37
6330	6664.872917	6630.728011	34.144906	0.51
6340	6673.539418	6630.728011	42.811407	0.64
6350	6682.565640	6723.132470	40.566830	0.61
6360	6691.704920	6723.132470	31.427550	0.47
6370	6700.871368	6723.132470	22.261102	0.33
6380	6709.726992	6723.132470	13.405479	0.20
6390	6718.186520	6726.740534	8.554014	0.13
6400	6727.861353	6727.737349	0.124004	0.00
6410	6736.409605	6736.249665	0.159941	0.00
6420	6745.670850	6745.569684	0.101166	0.00
6430	6754.800870	6755.325291	0.524421	0.01
6440	6763.550805	6758.104465	5.446340	0.08
6450	6772.495693	6758.104465	14.391228	0.21
6460	6781.893988	6758.104465	23.789523	0.35
6470	6791.028915	6758.104465	32.924451	0.48
6480	6799.273322	6758.104465	41.168857	0.61
6490	6808.690302	6838.756476	30.066174	0.44
6500	6817.957478	6838.756476	20.798997	0.31

6510	6826.694559	6838.756476	12.061917	0.18
6520	6835.801628	6838.756476	2.954847	0.04
6530	6844.542069	6838.756476	5.785594	0.08
6540	6853.366485	6838.756476	14.610009	0.21
6550	6862.251552	6838.756476	23.495077	0.34
6560	6871.789074	6904.195401	32.406327	0.47
6570	6880.377942	6904.195401	23.817459	0.35
6580	6889.289681	6904.195401	14.905720	0.22
6590	6898.502343	6904.195401	5.693058	0.08
6600	6907.140033	6906.026778	1.113254	0.02
6610	6916.245064	6906.026778	10.218285	0.15
6620	6924.460428	6924.251872	0.208556	0.00
6630	6934.184536	6924.251872	9.932664	0.14
6640	6942.947799	6958.552272	15.604474	0.22
6650	6951.731757	6958.552272	6.820516	0.10
6660	6960.901480	6958.552272	2.349207	0.03
6670	6969.932238	6958.552272	11.379965	0.16
6680	6978.842393	6985.973329	7.130936	0.10
6690	6987.254126	6985.973329	1.280798	0.02
6700	6996.886976	7010.537766	13.650790	0.20
6710	7005.100565	7010.537766	5.437201	0.08
6720	7014.629880	7012.658526	1.971355	0.03
6730	7024.042832	7024.851997	0.809165	0.01
6740	7032.326789	7030.877751	1.449039	0.02
6750	7041.380467	7050.059710	8.679243	0.12
6760	7050.297213	7050.059710	0.237503	0.00
6770	7059.889597	7050.059710	9.829887	0.14
6780	7068.407665	7050.059710	18.347954	0.26
6790	7077.065912	7050.059710	27.006202	0.38
6800	7086.306731	7105.065437	18.758706	0.26
6810	7095.085720	7105.065437	9.979717	0.14
6820	7104.510386	7105.065437	0.555051	0.01
6830	7112.700332	7123.779518	11.079186	0.16
6840	7122.205852	7123.779518	1.573666	0.02
6850	7130.710717	7128.202647	2.508070	0.04
6860	7139.857139	7128.202647	11.654492	0.16
6870	7149.102831	7128.202647	20.900184	0.29
6880	7157.560009	7128.202647	29.357362	0.41
6890	7166.532447	7199.528386	32.995939	0.46
6900	7175.547479	7199.528386	23.980908	0.33
6910	7184.720860	7199.528386	14.807527	0.21
6920	7193.221050	7199.528386	6.307336	0.09
6930	7202.481415	7202.718313	0.236898	0.00
6940	7211.335323	7203.097259	8.238065	0.11
6950	7220.432148	7203.097259	17.334889	0.24
6960	7228.963985	7203.097259	25.866726	0.36
6970	7237.923816	7203.097259	34.826557	0.48
6980	7247.172743	7203.097259	44.075484	0.61
6990	7255.894616	7291.168511	35.273895	0.49
7000	7264.748248	7291.168511	26.420263	0.36
7010	7273.701234	7291.168511	17.467277	0.24
7020	7282.817682	7291.168511	8.350829	0.11
7030	7291.290044	7291.168511	0.121533	0.00
7040	7300.763477	7291.168511	9.594966	0.13
7050	7309.497904	7291.168511	18.329393	0.25
7060	7318.351003	7291.168511	27.182491	0.37

7070	7327.534498	7354.736725	27.202227	0.37
7080	7336.016820	7354.736725	18.719905	0.26
7090	7345.413384	7354.736725	9.323341	0.13
7100	7353.913948	7354.736725	0.822777	0.01
7110	7363.102591	7354.736725	8.365866	0.11
7120	7371.987961	7354.736725	17.251236	0.23
7130	7380.464012	7354.736725	25.727287	0.35
7140	7389.584509	7354.736725	34.847784	0.47
7150	7398.751921	7354.736725	44.015196	0.59
7160	7407.782773	7459.554094	51.771322	0.70
7170	7416.269504	7459.554094	43.284590	0.58
7180	7425.306206	7459.554094	34.247889	0.46
7190	7434.041575	7459.554094	25.512519	0.34
7200	7443.171161	7459.554094	16.382934	0.22
7210	7451.287377	7459.554094	8.266717	0.11
7220	7460.742319	7459.554094	1.188225	0.02
7230	7469.742313	7467.221764	2.520549	0.03
7240	7477.990220	7489.445885	11.455665	0.15
7250	7486.980575	7489.445885	2.465310	0.03
7260	7495.620536	7497.338128	1.717592	0.02
7270	7505.007628	7497.338128	7.669500	0.10
7280	7513.256479	7500.402090	12.854388	0.17
7290	7522.235956	7508.156650	14.079306	0.19
7300	7531.443038	7508.156650	23.286387	0.31
7310	7540.227540	7508.156650	32.070889	0.43
7320	7548.467009	7508.156650	40.310359	0.53
7330	7557.731650	7601.752902	44.021252	0.58
7340	7566.845322	7601.752902	34.907579	0.46
7350	7575.696668	7601.752902	26.056233	0.34
7360	7584.486960	7601.752902	17.265942	0.23
7370	7593.098314	7608.436884	15.338570	0.20
7380	7602.405163	7611.158082	8.752918	0.12
7390	7610.854490	7611.158082	0.303591	0.00
7400	7619.749442	7611.158082	8.591360	0.11
7410	7628.742026	7611.158082	17.583945	0.23
7420	7637.940366	7657.227102	19.286735	0.25
7430	7646.560906	7657.227102	10.666196	0.14
7440	7655.121495	7657.227102	2.105607	0.03
7450	7664.203009	7657.227102	6.975908	0.09
7460	7673.103534	7669.444980	3.658554	0.05
7470	7681.945482	7687.416508	5.471026	0.07
7480	7690.429651	7687.416508	3.013142	0.04
7490	7700.036048	7687.416508	12.619539	0.16
7500	7708.220973	7731.909360	23.688387	0.31
7510	7717.430626	7731.909360	14.478734	0.19
7520	7725.881075	7731.909360	6.028285	0.08
7530	7734.799052	7734.338589	0.460463	0.01
7540	7744.168742	7738.779935	5.388807	0.07
7550	7752.412059	7738.779935	13.632124	0.18
7560	7761.852016	7744.990206	16.861811	0.22
7570	7770.261244	7744.990206	25.271038	0.33
7580	7779.147738	7744.990206	34.157532	0.44
7590	7787.847577	7818.139511	30.291934	0.39
7600	7796.882751	7818.139511	21.256761	0.27
7610	7805.651244	7818.139511	12.488267	0.16
7620	7814.556090	7826.564885	12.008795	0.15

7630	7822.719034	7826.564885	3.845851	0.05
7640	7832.274463	7829.036475	3.237988	0.04
7650	7840.782349	7829.036475	11.745874	0.15
7660	7849.322974	7865.170763	15.847789	0.20
7670	7857.998248	7865.170763	7.172515	0.09
7680	7867.142338	7865.170763	1.971575	0.03
7690	7876.081256	7870.596169	5.485087	0.07
7700	7884.386136	7870.596169	13.789967	0.17
7710	7893.753776	7870.596169	23.157607	0.29
7720	7902.431940	7870.596169	31.835771	0.40
7730	7911.290475	7945.968060	34.677586	0.44
7740	7919.702077	7945.968060	26.265983	0.33
7750	7928.684235	7945.968060	17.283826	0.22
7760	7937.745181	7945.968060	8.222880	0.10
7770	7946.220788	7945.968060	0.252728	0.00
7780	7954.747609	7952.425719	2.321890	0.03
7790	7964.203601	7952.425719	11.777882	0.15
7800	7972.995332	7952.425719	20.569613	0.26
7810	7981.677159	7952.425719	29.251440	0.37
7820	7990.464730	8013.609674	23.144944	0.29
7830	7998.960516	8013.609674	14.649158	0.18
7840	8008.486686	8013.609674	5.122988	0.06
7850	8016.491200	8013.609674	2.881526	0.04
7860	8025.752453	8023.925870	1.826583	0.02
7870	8034.386241	8023.925870	10.460371	0.13
7880	8043.228212	8023.925870	19.302342	0.24
7890	8051.826701	8023.925870	27.900831	0.35
7900	8060.656751	8081.910634	21.253883	0.26
7910	8069.620474	8081.910634	12.290161	0.15
7920	8078.331291	8081.910634	3.579343	0.04
7930	8087.201333	8081.910634	5.290699	0.07
7940	8095.644650	8098.111906	2.467256	0.03
7950	8104.959581	8120.973090	16.013510	0.20
7960	8113.552643	8120.973090	7.420447	0.09
7970	8122.361423	8120.973090	1.388333	0.02
7980	8130.717736	8120.973090	9.744645	0.12
7990	8139.566538	8157.929523	18.362985	0.23
8000	8148.188406	8157.929523	9.741117	0.12
8010	8157.194765	8157.929523	0.734758	0.01
8020	8165.448818	8159.679519	5.769300	0.07
8030	8174.366181	8166.844756	7.521425	0.09
8040	8183.129990	8166.844756	16.285235	0.20
8050	8192.334650	8197.811283	5.476633	0.07
8060	8200.341952	8197.811283	2.530669	0.03
8070	8209.530686	8225.908481	16.377795	0.20
8080	8218.619573	8225.908481	7.288907	0.09
8090	8227.022032	8225.908481	1.113552	0.01
8100	8235.903078	8228.353788	7.549290	0.09
8110	8244.522473	8249.533524	5.011051	0.06
8120	8253.795262	8249.533524	4.261738	0.05
8130	8262.002122	8261.035026	0.967097	0.01
8140	8271.102989	8273.695643	2.592654	0.03
8150	8279.658220	8280.834024	1.175804	0.01
8160	8288.980436	8284.958530	4.021906	0.05
8170	8297.216961	8284.958530	12.258431	0.15
8180	8306.142916	8284.958530	21.184387	0.26

8190	8314.936715	8284.958530	29.978185	0.36
8200	8323.508788	8284.958530	38.550259	0.46
8210	8332.263071	8366.254430	33.991359	0.41
8220	8341.281555	8366.254430	24.972875	0.30
8230	8349.534979	8366.254430	16.719451	0.20
8240	8358.619426	8366.254430	7.635004	0.09
8250	8367.570203	8366.254430	1.315773	0.02
8260	8375.843575	8366.254430	9.589145	0.11
8270	8384.243265	8374.448808	9.794457	0.12
8280	8393.253383	8374.448808	18.804575	0.22
8290	8402.071249	8374.448808	27.622441	0.33
8300	8410.365241	8374.448808	35.916433	0.43
8310	8419.569109	8454.243653	34.674544	0.41
8320	8428.018573	8454.243653	26.225080	0.31
8330	8437.043924	8456.672030	19.628106	0.23
8340	8445.343743	8459.976583	14.632840	0.17
8350	8454.385594	8462.638102	8.252508	0.10
8360	8462.646579	8462.638102	0.008477	0.00
8370	8472.106757	8475.317209	3.210453	0.04
8380	8480.498098	8478.081819	2.416279	0.03
8390	8489.335844	8497.471233	8.135390	0.10
8400	8497.637356	8497.471233	0.166123	0.00
8410	8506.655630	8502.405844	4.249786	0.05
8420	8515.510306	8502.405844	13.104462	0.15
8430	8524.072960	8505.751702	18.321258	0.21
8440	8532.610707	8508.759511	23.851196	0.28
8450	8541.539983	8511.214282	30.325701	0.36
8460	8550.742265	8511.214282	39.527983	0.46
8470	8558.923461	8511.214282	47.709179	0.56
8480	8567.744540	8601.710652	33.966112	0.40
8490	8576.708935	8601.710652	25.001717	0.29
8500	8585.651463	8601.710652	16.059189	0.19
8510	8594.055749	8604.966693	10.910945	0.13
8520	8602.800931	8611.124924	8.323994	0.10
8530	8610.900549	8611.124924	0.224375	0.00
8540	8619.610075	8611.124924	8.485151	0.10
8550	8628.431748	8611.124924	17.306824	0.20
8560	8637.187564	8611.124924	26.062640	0.30
8570	8645.629539	8669.263466	23.633927	0.27
8580	8654.501099	8669.263466	14.762367	0.17
8590	8663.160875	8669.263466	6.102591	0.07
8600	8671.820092	8669.263466	2.556625	0.03
8610	8680.584051	8669.263466	11.320585	0.13
8620	8689.551335	8669.263466	20.287869	0.23
8630	8697.610486	8669.263466	28.347020	0.33
8640	8707.005237	8669.263466	37.741770	0.43
8650	8715.585780	8669.263466	46.322314	0.53
8660	8724.108198	8781.133174	57.024976	0.65
8670	8732.623378	8781.133174	48.509797	0.56
8680	8741.683326	8781.133174	39.449848	0.45
8690	8750.330087	8781.133174	30.803088	0.35
8700	8758.806686	8781.133174	22.326488	0.25
8710	8767.797893	8781.133174	13.335281	0.15
8720	8776.412827	8781.133174	4.720347	0.05
8730	8785.349006	8784.410984	0.938022	0.01
8740	8793.748088	8786.035557	7.712530	0.09

8750	8802.633958	8800.276898	2.357060	0.03
8760	8810.602647	8800.276898	10.325750	0.12
8770	8819.823257	8823.910530	4.087272	0.05
8780	8828.126900	8823.910530	4.216370	0.05
8790	8836.623445	8823.910530	12.712916	0.14
8800	8845.399046	8823.910530	21.488516	0.24
8810	8854.558251	8823.910530	30.647722	0.35
8820	8862.808651	8892.218248	29.409597	0.33
8830	8871.689921	8892.218248	20.528327	0.23
8840	8880.343782	8892.218248	11.874466	0.13
8850	8888.831197	8892.218248	3.387051	0.04
8860	8897.673494	8897.633141	0.040353	0.00
8870	8906.334723	8897.633141	8.701582	0.10
8880	8914.741189	8924.148150	9.406961	0.11
8890	8923.430369	8924.148150	0.717781	0.01
8900	8932.151532	8934.365271	2.213739	0.02
8910	8941.181572	8944.948364	3.766792	0.04
8920	8949.659175	8950.388484	0.729309	0.01
8930	8958.329613	8950.388484	7.941129	0.09
8940	8966.831853	8950.388484	16.443368	0.18
8950	8976.046813	8950.388484	25.658329	0.29
8960	8984.458470	8950.388484	34.069986	0.38
8970	8993.206523	8950.388484	42.818039	0.48
8980	9001.129405	9038.967005	37.837600	0.42
8990	9010.088963	9038.967005	28.878043	0.32
9000	9018.424476	9038.967005	20.542530	0.23
9010	9027.231505	9038.967005	11.735501	0.13
9020	9035.749059	9038.967005	3.217946	0.04
9030	9044.618071	9042.730949	1.887122	0.02
9040	9052.755417	9042.730949	10.024468	0.11
9050	9062.010375	9066.229581	4.219206	0.05
9060	9070.419821	9067.483190	2.936631	0.03
9070	9078.885069	9068.560823	10.324245	0.11
9080	9087.619989	9068.560823	19.059166	0.21
9090	9096.611589	9068.560823	28.050766	0.31
9100	9105.074921	9115.398883	10.323962	0.11
9110	9113.658797	9115.398883	1.740086	0.02
9120	9122.504965	9115.398883	7.106082	0.08
9130	9130.768159	9117.562119	13.206040	0.14
9140	9140.231970	9117.562119	22.669851	0.25
9150	9148.555047	9181.995097	33.440050	0.37
9160	9157.200854	9181.995097	24.794243	0.27
9170	9165.448663	9181.995097	16.546433	0.18
9180	9174.799448	9186.463071	11.663622	0.13
9190	9182.811174	9186.463071	3.651897	0.04
9200	9191.463119	9189.290078	2.173041	0.02
9210	9200.006746	9205.537613	5.530867	0.06
9220	9208.520414	9209.632866	1.112451	0.01
9230	9216.889163	9212.041616	4.847547	0.05
9240	9226.173216	9213.929921	12.243295	0.13
9250	9234.086402	9215.613410	18.472991	0.20
9260	9243.002727	9215.613410	27.389317	0.30
9270	9251.325332	9270.519252	19.193920	0.21
9280	9260.462686	9270.519252	10.056565	0.11
9290	9268.846664	9274.454984	5.608321	0.06
9300	9277.608768	9274.454984	3.153784	0.03

9310	9285.849431	9274.454984	11.394447	0.12
9320	9294.994414	9274.454984	20.539430	0.22
9330	9303.238853	9322.203113	18.964260	0.20
9340	9312.287090	9322.203113	9.916023	0.11
9350	9320.592913	9322.203113	1.610199	0.02
9360	9329.235940	9322.203113	7.032828	0.08
9370	9337.535506	9322.203113	15.332393	0.16
9380	9346.909626	9375.692357	28.782731	0.31
9390	9355.235106	9375.692357	20.457251	0.22
9400	9363.690017	9375.692357	12.002340	0.13
9410	9371.587600	9375.692357	4.104757	0.04
9420	9380.793874	9375.692357	5.101517	0.05
9430	9388.979170	9375.692357	13.286813	0.14
9440	9397.818268	9398.138452	0.320185	0.00
9450	9406.317820	9398.138452	8.179368	0.09
9460	9415.106792	9398.138452	16.968340	0.18
9470	9423.591009	9436.337502	12.746493	0.14
9480	9432.713826	9436.337502	3.623676	0.04
9490	9440.804070	9440.506792	0.297278	0.00
9500	9449.609725	9444.147500	5.462226	0.06
9510	9457.972397	9452.544896	5.427501	0.06
9520	9466.722587	9452.544896	14.177691	0.15
9530	9475.565521	9492.994751	17.429230	0.18
9540	9484.207298	9492.994751	8.787453	0.09
9550	9492.559852	9492.994751	0.434900	0.00
9560	9501.217268	9498.906782	2.310486	0.02
9570	9509.769914	9513.212619	3.442705	0.04
9580	9518.401462	9514.034931	4.366532	0.05
9590	9526.527981	9514.034931	12.493051	0.13
9600	9535.377270	9548.355187	12.977918	0.14
9610	9543.327342	9548.355187	5.027845	0.05
9620	9552.369737	9548.355187	4.014550	0.04
9630	9560.804276	9548.355187	12.449089	0.13
9640	9569.757584	9580.008665	10.251081	0.11
9650	9578.082269	9580.008665	1.926396	0.02
9660	9586.572222	9588.965282	2.393060	0.02
9670	9594.988286	9590.850075	4.138211	0.04
9680	9604.113456	9590.850075	13.263381	0.14
9690	9612.606350	9618.017665	5.411315	0.06
9700	9621.165759	9618.017665	3.148094	0.03
9710	9629.724366	9630.759706	1.035340	0.01
9720	9638.333970	9630.759706	7.574264	0.08
9730	9646.682120	9654.396531	7.714411	0.08
9740	9655.670243	9654.727701	0.942542	0.01
9750	9663.881077	9659.102753	4.778323	0.05
9760	9672.494340	9659.102753	13.391587	0.14
9770	9680.762834	9679.594751	1.168083	0.01
9780	9689.424900	9698.705682	9.280782	0.10
9790	9697.964079	9698.705682	0.741604	0.01
9800	9706.802866	9705.537947	1.264919	0.01
9810	9714.689106	9705.537947	9.151159	0.09
9820	9723.737313	9721.864830	1.872483	0.02
9830	9732.006894	9731.034875	0.972018	0.01
9840	9741.076088	9737.109658	3.966430	0.04
9850	9749.482254	9750.278165	0.795911	0.01
9860	9758.307454	9762.525633	4.218178	0.04

```
9870  9766.238755  9763.682307     2.556448            0.03
9880  9775.088284  9767.169862     7.918422            0.08
9890  9783.722535  9785.227369     1.504835            0.02
9900  9792.246729  9794.121353     1.874624            0.02
9910  9800.831815  9800.107953     0.723862            0.01
9920  9809.662905  9805.511479     4.151427            0.04
9930  9817.665816  9818.357695     0.691879            0.01
9940  9826.946695  9825.779708     1.166988            0.01
9950  9834.670765  9830.832227     3.838538            0.04
9960  9843.426279  9838.054125     5.372154            0.05
9970  9851.623957  9848.818224     2.805733            0.03
9980  9860.360205  9855.559667     4.800538            0.05
9990  9868.714659  9862.647440     6.067220            0.06
10000  9877.782654  9864.045983    13.736671            0.14
------------------------------------------------------------------------
--------
Mean absolute difference: 14.649778
Mean relative error: 0.52%
Max difference: 62.398698

CONSERVATIVE MODEL:
========================================================================
========
  #       Theory     Computed         Diff    Rel.Error(%)
------------------------------------------------------------------------
--------
  1    14.134725  2196.840697  2182.705972        15442.15
  2    21.022040  2196.862407  2175.840368        10350.28
  3    25.010858  2196.874749  2171.863891         8683.68
  4    30.424876  2196.893050  2166.468174         7120.71
  5    32.935062  2196.925519  2163.990458         6570.48
  6    37.586178  2196.932562  2159.346384         5745.05
  7    40.918719  2196.974245  2156.055526         5269.12
  8    43.327073  2197.005987  2153.678914         4970.75
  9    48.005151  2197.022178  2149.017027         4476.64
 10    49.773832  2197.067513  2147.293680         4314.10
 20    77.144840  2197.332289  2120.187449         2748.32
 30   101.317851  2197.659275  2096.341424         2069.07
 40   122.946829  2198.014065  2075.067236         1687.78
 50   143.111846  2198.369924  2055.258078         1436.12
 60   163.030710  2198.746123  2035.715414         1248.67
 70   182.207078  2199.225072  2017.017994         1106.99
 80   201.264752  2199.562241  1998.297489          992.87
 90   219.067596  2200.034887  1980.967291          904.27
100   236.524230  2200.515435  1963.991206          830.36
110   253.069987  2201.013852  1947.943865          769.73
120   269.970449  2201.424852  1931.454403          715.43
130   286.667445  2202.094595  1915.427150          668.17
140   302.696750  2202.625693  1899.928943          627.67
150   318.853104  2203.217488  1884.364384          590.98
160   334.211355  2203.712370  1869.501015          559.38
170   350.408419  2204.359114  1853.950694          529.08
180   366.212710  2204.900398  1838.687688          502.08
190   381.484469  2205.613350  1824.128882          478.17
200   396.381854  2206.228789  1809.846935          456.59
210   411.972268  2206.860157  1794.887889          435.68
```

220	427.208825	2207.426776	1780.217951	416.71
230	441.683199	2208.034591	1766.351392	399.91
240	456.328427	2208.564112	1752.235685	383.99
250	470.773655	2209.070437	1738.296782	369.24
260	485.539148	2209.524415	1723.985267	355.07
270	500.309085	2209.902700	1709.593615	341.71
280	513.668986	2210.342420	1696.673434	330.30
290	528.406214	2210.820606	1682.414392	318.39
300	541.847437	2211.387528	1669.540090	308.12
310	556.899476	2211.969958	1655.070481	297.19
320	570.051115	2212.579085	1642.527970	288.14
330	584.561706	2213.181440	1628.619734	278.61
340	598.493077	2213.874818	1615.381740	269.91
350	611.774210	2214.595120	1602.820911	262.00
360	626.019283	2215.338637	1589.319353	253.88
370	638.927938	2216.002049	1577.074111	246.83
380	653.649572	2216.842902	1563.193330	239.15
390	666.515148	2217.660856	1551.145709	232.72
400	679.742198	2218.475150	1538.732952	226.37
410	693.176970	2219.223559	1526.046589	220.15
420	706.184655	2220.168201	1513.983546	214.39
430	719.697101	2221.018255	1501.321154	208.60
440	732.818053	2221.923975	1489.105922	203.20
450	746.499306	2222.894467	1476.395161	197.78
460	758.900238	2223.766295	1464.866057	193.02
470	772.961618	2224.729580	1451.767962	187.82
480	785.739090	2225.763346	1440.024257	183.27
490	798.707570	2226.787493	1428.079922	178.80
500	811.184359	2227.840902	1416.656543	174.64
510	824.526294	2228.878582	1404.352288	170.32
520	837.347335	2229.975212	1392.627876	166.31
530	849.862274	2231.132897	1381.270623	162.53
540	863.189720	2232.247144	1369.057424	158.60
550	875.985285	2233.371729	1357.386444	154.96
560	888.475567	2234.575897	1346.100330	151.51
570	900.849740	2235.772293	1334.922553	148.18
580	912.823999	2236.974730	1324.150731	145.06
590	926.551553	2238.200337	1311.648784	141.56
600	939.024301	2239.368898	1300.344597	138.48
610	951.033249	2240.728752	1289.695503	135.61
620	963.567040	2241.994813	1278.427773	132.68
630	976.178502	2243.328119	1267.149617	129.81
640	988.992622	2244.615151	1255.622529	126.96
650	1001.349483	2246.084202	1244.734719	124.31
660	1013.058638	2247.435924	1234.377286	121.85
670	1025.707944	2248.923066	1223.215122	119.26
680	1038.087752	2250.337165	1212.249413	116.78
690	1049.996284	2251.768460	1201.772175	114.45
700	1062.915382	2253.235051	1190.319669	111.99
710	1074.747771	2254.616722	1179.868951	109.78
720	1086.911999	2256.218299	1169.306300	107.58
730	1099.360667	2257.742768	1158.382101	105.37
740	1111.443505	2259.266009	1147.822505	103.27
750	1123.101117	2260.859958	1137.758841	101.31
760	1135.562214	2262.449477	1126.887263	99.24
770	1147.501777	2264.088386	1116.586610	97.31

780	1159.480657	2265.753028	1106.272371	95.41
790	1172.120682	2267.373160	1095.252478	93.44
800	1183.712775	2269.112143	1085.399368	91.69
810	1196.034672	2270.816924	1074.782252	89.86
820	1208.471460	2272.497349	1064.025889	88.05
830	1219.614471	2274.255877	1054.641406	86.47
840	1231.562274	2276.041931	1044.479657	84.81
850	1243.538147	2277.864810	1034.326664	83.18
860	1255.408231	2279.637186	1024.228955	81.59
870	1267.200346	2281.436277	1014.235932	80.04
880	1279.332843	2283.300246	1003.967403	78.48
890	1290.417708	2285.111927	994.694219	77.08
900	1302.346742	2287.042570	984.695828	75.61
910	1314.052566	2288.948728	974.896163	74.19
920	1325.981970	2290.941923	964.959953	72.77
930	1337.688792	2292.851699	955.162908	71.40
940	1349.085194	2294.844202	945.759008	70.10
950	1361.393075	2296.888553	935.495479	68.72
960	1373.202915	2298.911740	925.708825	67.41
970	1384.444416	2300.959910	916.515494	66.20
980	1396.544163	2302.978843	906.434680	64.91
990	1408.136307	2305.004247	896.867939	63.69
1000	1419.422481	2307.145671	887.723190	62.54
1010	1431.218314	2309.305604	878.087291	61.35
1020	1442.911060	2311.476639	868.565580	60.20
1030	1454.226011	2313.624311	859.398300	59.10
1040	1465.889558	2315.822880	849.933323	57.98
1050	1477.435191	2318.013354	840.578163	56.89
1060	1489.252467	2320.270115	831.017648	55.80
1070	1500.351952	2322.509850	822.157898	54.80
1080	1511.923005	2324.795583	812.872578	53.76
1090	1523.486047	2327.099684	803.613637	52.75
1100	1535.042277	2329.334579	794.292302	51.74
1110	1546.091198	2331.718622	785.627424	50.81
1120	1557.394507	2334.057959	776.663452	49.87
1130	1569.022478	2336.418123	767.395645	48.91
1140	1580.372082	2338.814170	758.442088	47.99
1150	1591.293884	2341.285876	749.991992	47.13
1160	1602.815405	2343.725393	740.909989	46.23
1170	1614.710628	2346.150031	731.439404	45.30
1180	1625.987020	2348.652980	722.665960	44.44
1190	1637.089190	2351.121346	714.032156	43.62
1200	1648.270598	2353.700084	705.429486	42.80
1210	1660.019980	2356.228028	696.208048	41.94
1220	1671.047182	2358.752954	687.705772	41.15
1230	1681.812320	2361.343471	679.531151	40.40
1240	1693.053021	2363.955839	670.902818	39.63
1250	1704.291412	2366.667784	662.376371	38.87
1260	1715.825283	2369.291181	653.465898	38.08
1270	1726.605701	2371.945869	645.340168	37.38
1280	1737.893885	2374.700778	636.806894	36.64
1290	1748.563987	2377.364656	628.800669	35.96
1300	1760.215166	2380.083737	619.868571	35.22
1310	1771.212945	2382.846679	611.633734	34.53
1320	1782.439067	2385.521070	603.082003	33.83
1330	1793.525391	2388.433868	594.908477	33.17

1340	1804.904161	2391.190108	586.285947	32.48
1350	1816.338611	2394.093752	577.755141	31.81
1360	1826.795455	2396.941072	570.145617	31.21
1370	1838.086943	2399.764525	561.677582	30.56
1380	1848.722761	2402.675938	553.953177	29.96
1390	1860.541388	2405.608302	545.066914	29.30
1400	1871.416043	2408.578491	537.162448	28.70
1410	1882.321975	2411.514391	529.192416	28.11
1420	1892.867701	2414.513554	521.645853	27.56
1430	1904.089325	2417.466508	513.377182	26.96
1440	1915.054629	2420.561343	505.506714	26.40
1450	1926.211583	2423.569185	497.357602	25.82
1460	1936.873094	2426.608144	489.735050	25.28
1470	1947.917048	2429.687282	481.770235	24.73
1480	1958.851883	2432.837278	473.985395	24.20
1490	1970.305010	2435.978490	465.673481	23.63
1500	1980.910043	2439.143585	458.233541	23.13
1510	1992.001706	2442.294584	450.292878	22.61
1520	2002.896408	2445.433011	442.536602	22.09
1530	2013.872098	2448.644625	434.772527	21.59
1540	2024.488262	2451.954331	427.466069	21.11
1550	2035.077440	2455.181686	420.104246	20.64
1560	2045.802760	2458.435946	412.633186	20.17
1570	2056.620867	2461.649210	405.028344	19.69
1580	2068.345639	2465.014531	396.668893	19.18
1590	2078.939288	2468.392041	389.452753	18.73
1600	2090.036207	2471.763541	381.727334	18.26
1610	2100.593745	2475.074369	374.480623	17.83
1620	2110.822372	2478.477333	367.654961	17.42
1630	2122.465151	2481.861816	359.396664	16.93
1640	2132.798516	2485.321393	352.522876	16.53
1650	2143.374444	2488.795472	345.421027	16.12
1660	2154.263624	2492.278178	338.014554	15.69
1670	2165.011437	2495.701130	330.689692	15.27
1680	2176.204119	2499.251526	323.047406	14.84
1690	2187.085691	2502.863767	315.778076	14.44
1700	2197.258747	2506.327590	309.068843	14.07
1710	2207.802741	2509.982163	302.179421	13.69
1720	2218.521007	2513.579557	295.058550	13.30
1730	2229.127959	2517.196951	288.068993	12.92
1740	2240.021209	2520.848063	280.826854	12.54
1750	2251.046642	2524.549238	273.502596	12.15
1760	2261.660064	2528.156940	266.496876	11.78
1770	2272.049016	2531.923080	259.874064	11.44
1780	2282.690375	2535.606464	252.916089	11.08
1790	2293.896145	2539.226686	245.330540	10.69
1800	2304.305139	2543.201971	238.896832	10.37
1810	2314.621182	2546.824343	232.203161	10.03
1820	2325.557096	2550.722838	225.165742	9.68
1830	2335.867154	2554.599409	218.732255	9.36
1840	2346.464120	2558.384404	211.920284	9.03
1850	2357.608316	2562.315003	204.706687	8.68
1860	2367.957702	2566.192208	198.234507	8.37
1870	2378.212962	2570.045938	191.832977	8.07
1880	2388.748208	2573.971714	185.223506	7.75
1890	2399.433833	2577.926858	178.493025	7.44

1900	2409.881603	2581.829074	171.947471	7.14
1910	2421.142944	2585.933527	164.790583	6.81
1920	2431.611135	2589.934404	158.323269	6.51
1930	2441.929837	2593.862173	151.932336	6.22
1940	2452.283366	2597.985835	145.702469	5.94
1950	2462.464295	2602.038658	139.574363	5.67
1960	2473.107125	2606.138875	133.031750	5.38
1970	2484.503238	2610.238304	125.735066	5.06
1980	2494.866178	2614.338343	119.472165	4.79
1990	2504.772153	2618.525926	113.753774	4.54
2000	2515.286483	2622.711395	107.424912	4.27
2010	2525.687993	2626.881543	101.193549	4.01
2020	2536.091885	2631.077868	94.985983	3.75
2030	2546.761141	2635.346551	88.585410	3.48
2040	2558.009106	2639.570778	81.561672	3.19
2050	2567.958600	2643.848519	75.889919	2.96
2060	2578.070130	2648.127926	70.057796	2.72
2070	2588.368210	2652.448454	64.080244	2.48
2080	2598.598768	2656.770999	58.172231	2.24
2090	2609.552087	2661.105997	51.553910	1.98
2100	2620.279394	2665.519562	45.240169	1.73
2110	2629.994801	2669.860106	39.865305	1.52
2120	2640.462943	2674.295724	33.832781	1.28
2130	2651.227331	2678.801572	27.574241	1.04
2140	2661.582574	2683.202835	21.620262	0.81
2150	2671.918383	2687.719287	15.800904	0.59
2160	2681.841051	2692.139428	10.298376	0.38
2170	2692.368719	2696.678705	4.309986	0.16
2180	2703.143658	2701.178738	1.964920	0.07
2190	2713.525289	2705.758256	7.767033	0.29
2200	2723.768929	2710.294746	13.474183	0.49
2210	2734.200783	2715.019628	19.181155	0.70
2220	2743.953470	2719.571434	24.382035	0.89
2230	2754.458663	2724.187970	30.270693	1.10
2240	2765.407691	2728.850993	36.556699	1.32
2250	2775.725707	2733.477403	42.248304	1.52
2260	2786.043336	2738.206189	47.837146	1.72
2270	2796.034762	2742.929682	53.105079	1.90
2280	2806.002732	2747.638755	58.363977	2.08
2290	2816.337955	2752.406852	63.931103	2.27
2300	2826.712170	2757.127022	69.585148	2.46
2310	2837.014472	2761.956553	75.057919	2.65
2320	2847.535936	2766.848071	80.687864	2.83
2330	2857.736140	2771.507747	86.228393	3.02
2340	2868.224164	2776.523719	91.700445	3.20
2350	2878.527003	2781.373183	97.153819	3.38
2360	2888.394382	2786.162035	102.232348	3.54
2370	2898.541353	2791.136185	107.405168	3.71
2380	2908.658046	2796.082206	112.575840	3.87
2390	2919.321772	2801.026142	118.295631	4.05
2400	2930.004366	2805.952045	124.052321	4.23
2410	2939.847529	2810.961418	128.886111	4.38
2420	2949.790143	2815.991440	133.798703	4.54
2430	2960.033618	2821.028541	139.005077	4.70
2440	2970.049930	2826.120200	143.929730	4.85
2450	2980.487381	2831.147821	149.339560	5.01

2460	2990.423271	2836.248574	154.174697	5.16
2470	3001.023331	2841.346337	159.676994	5.32
2480	3011.344380	2846.478795	164.865585	5.47
2490	3021.420401	2851.636549	169.783852	5.62
2500	3031.289217	2856.780882	174.508335	5.76
2510	3041.256804	2861.961628	179.295176	5.90
2520	3051.342383	2867.174887	184.167497	6.04
2530	3061.837584	2872.419735	189.417849	6.19
2540	3072.000993	2877.635924	194.365069	6.33
2550	3082.163164	2882.911561	199.251603	6.46
2560	3092.687667	2888.219526	204.468141	6.61
2570	3102.436089	2893.484374	208.951715	6.74
2580	3112.840738	2898.861959	213.978779	6.87
2590	3122.610129	2904.103551	218.506577	7.00
2600	3132.698331	2909.521735	223.176596	7.12
2610	3142.979457	2914.967725	228.011731	7.25
2620	3152.994560	2920.259565	232.734995	7.38
2630	3163.311960	2925.744955	237.567004	7.51
2640	3173.011634	2931.113418	241.898216	7.62
2650	3183.205673	2936.578300	246.627374	7.75
2660	3193.557208	2942.106875	251.450334	7.87
2670	3203.630891	2947.637238	255.993653	7.99
2680	3213.606499	2953.051376	260.555123	8.11
2690	3223.372566	2958.752685	264.619881	8.21
2700	3233.913361	2964.237016	269.676345	8.34
2710	3243.501653	2969.841754	273.659898	8.44
2720	3253.369017	2975.432858	277.936160	8.54
2730	3263.784940	2981.090282	282.694658	8.66
2740	3274.059953	2986.668376	287.391577	8.78
2750	3284.499863	2992.324520	292.175343	8.90
2760	3293.975869	2998.004544	295.971325	8.99
2770	3303.813719	3003.720230	300.093489	9.08
2780	3313.842662	3009.418188	304.424474	9.19
2790	3323.875125	3015.120335	308.754790	9.29
2800	3333.781383	3020.883950	312.897433	9.39
2810	3343.604909	3026.587879	317.017029	9.48
2820	3354.278231	3032.481911	321.796321	9.59
2830	3364.355764	3038.294429	326.061334	9.69
2840	3374.427472	3044.075050	330.352422	9.79
2850	3383.871212	3049.920660	333.950552	9.87
2860	3394.137666	3055.706093	338.431573	9.97
2870	3404.362762	3061.707576	342.655186	10.07
2880	3413.710819	3067.600143	346.110677	10.14
2890	3423.611395	3073.458432	350.152963	10.23
2900	3433.769710	3079.481885	354.287825	10.32
2910	3444.036571	3085.354996	358.681574	10.41
2920	3453.677894	3091.363327	362.314567	10.49
2930	3464.083832	3097.426751	366.657081	10.58
2940	3473.814882	3103.411352	370.403530	10.66
2950	3484.159388	3109.414914	374.744474	10.76
2960	3493.720613	3115.600519	378.120094	10.82
2970	3503.636216	3121.625679	382.010537	10.90
2980	3513.877574	3127.687008	386.190566	10.99
2990	3523.635836	3133.863556	389.772280	11.06
3000	3533.328243	3139.841485	393.486759	11.14
3010	3543.322307	3146.076422	397.245885	11.21

3020	3553.840681	3152.181719	401.658962	11.30
3030	3563.215121	3158.387551	404.827571	11.36
3040	3573.079804	3164.725442	408.354363	11.43
3050	3582.923042	3170.891385	412.031657	11.50
3060	3593.028559	3177.143416	415.885142	11.57
3070	3602.418894	3183.403181	419.015714	11.63
3080	3612.324193	3189.669695	422.654498	11.70
3090	3622.179275	3195.943715	426.235559	11.77
3100	3632.478178	3202.345683	430.132495	11.84
3110	3641.707123	3208.627383	433.079741	11.89
3120	3651.783523	3215.071022	436.712502	11.96
3130	3662.368949	3221.349853	441.019096	12.04
3140	3672.349351	3227.753865	444.595487	12.11
3150	3682.418731	3234.231550	448.187181	12.17
3160	3691.756940	3240.602165	451.154774	12.22
3170	3702.050757	3247.082453	454.968304	12.29
3180	3711.404818	3253.560118	457.844700	12.34
3190	3721.296040	3260.025831	461.270209	12.40
3200	3731.075159	3266.600726	464.474433	12.45
3210	3740.875154	3273.128832	467.746322	12.50
3220	3750.303158	3279.659440	470.643718	12.55
3230	3760.582729	3286.178611	474.404118	12.62
3240	3770.105498	3292.815661	477.289836	12.66
3250	3780.226486	3299.369188	480.857299	12.72
3260	3790.041777	3305.974472	484.067305	12.77
3270	3800.132426	3312.666601	487.465825	12.83
3280	3809.874248	3319.325631	490.548617	12.88
3290	3819.414499	3325.946936	493.467563	12.92
3300	3829.144810	3332.680559	496.464252	12.97
3310	3839.426268	3339.424322	500.001946	13.02
3320	3848.415301	3346.106623	502.308678	13.05
3330	3858.448586	3352.889574	505.559012	13.10
3340	3868.412859	3359.661774	508.751085	13.15
3350	3878.256175	3366.365490	511.890685	13.20
3360	3887.352014	3373.265682	514.086331	13.22
3370	3897.870750	3380.099981	517.770769	13.28
3380	3908.188165	3386.893924	521.294241	13.34
3390	3917.658166	3393.761099	523.897067	13.37
3400	3927.080517	3400.568866	526.511651	13.41
3410	3936.974234	3407.493594	529.480641	13.45
3420	3946.878984	3414.415675	532.463308	13.49
3430	3956.420217	3421.368557	535.051660	13.52
3440	3966.138715	3428.425556	537.713159	13.56
3450	3976.066183	3435.333249	540.732934	13.60
3460	3985.098725	3442.295160	542.803565	13.62
3470	3995.168460	3449.340338	545.828121	13.66
3480	4004.806731	3456.264976	548.541755	13.70
3490	4014.657295	3463.420393	551.236901	13.73
3500	4023.905151	3470.403085	553.502067	13.76
3510	4034.520820	3477.456019	557.064802	13.81
3520	4044.322287	3484.575625	559.746662	13.84
3530	4053.566272	3491.690648	561.875624	13.86
3540	4063.157725	3498.824412	564.333314	13.89
3550	4073.126450	3506.021598	567.104852	13.92
3560	4082.588348	3513.186716	569.401632	13.95
3570	4092.297206	3520.299879	571.997328	13.98

3580	4102.037363	3527.626790	574.410572	14.00
3590	4111.859845	3534.740591	577.119255	14.04
3600	4121.348657	3541.992826	579.355831	14.06
3610	4131.136980	3549.323800	581.813180	14.08
3620	4140.892830	3556.505784	584.387046	14.11
3630	4150.445098	3563.873762	586.571336	14.13
3640	4159.921666	3571.141395	588.780271	14.15
3650	4170.482806	3578.437805	592.045001	14.20
3660	4179.574699	3585.851607	593.723092	14.21
3670	4189.615280	3593.154295	596.460985	14.24
3680	4199.335193	3600.591685	598.743508	14.26
3690	4208.734782	3608.058510	600.676272	14.27
3700	4218.230789	3615.449282	602.781506	14.29
3710	4227.984705	3622.843911	605.140794	14.31
3720	4237.343105	3630.301136	607.041969	14.33
3730	4247.050666	3637.820955	609.229711	14.34
3740	4256.903916	3645.315058	611.588859	14.37
3750	4266.175526	3652.751003	613.424524	14.38
3760	4275.863648	3660.316209	615.547439	14.40
3770	4285.629054	3667.853960	617.775094	14.42
3780	4295.348211	3675.347895	620.000317	14.43
3790	4304.617528	3683.018251	621.599277	14.44
3800	4314.799283	3690.557550	624.241733	14.47
3810	4324.546748	3698.167392	626.379356	14.48
3820	4334.020581	3705.890492	628.130089	14.49
3830	4343.563201	3713.400575	630.162627	14.51
3840	4353.700190	3721.136732	632.563458	14.53
3850	4362.761583	3728.874765	633.886818	14.53
3860	4372.227907	3736.507996	635.719911	14.54
3870	4382.125949	3744.227595	637.898355	14.56
3880	4391.021213	3752.011796	639.009417	14.55
3890	4401.175472	3759.643035	641.532436	14.58
3900	4410.491142	3767.539592	642.951550	14.58
3910	4419.922458	3775.309664	644.612794	14.58
3920	4429.488010	3783.031438	646.456572	14.59
3930	4439.557561	3790.963888	648.593672	14.61
3940	4448.301901	3798.727413	649.574488	14.60
3950	4458.188749	3806.677740	651.511009	14.61
3960	4467.739817	3814.572519	653.167298	14.62
3970	4477.015857	3822.379146	654.636711	14.62
3980	4487.084191	3830.310546	656.773646	14.64
3990	4496.893452	3838.262903	658.630548	14.65
4000	4506.311497	3846.203719	660.107778	14.65
4010	4516.089845	3854.186568	661.903278	14.66
4020	4525.790882	3862.105066	663.685816	14.66
4030	4534.935556	3870.131008	664.804548	14.66
4040	4544.720271	3878.160977	666.559294	14.67
4050	4554.197951	3886.075355	668.122596	14.67
4060	4563.416046	3894.243705	669.172342	14.66
4070	4573.246173	3902.267583	670.978591	14.67
4080	4582.568249	3910.342531	672.225718	14.67
4090	4592.049309	3918.474094	673.575215	14.67
4100	4601.602962	3926.555942	675.047019	14.67
4110	4611.484674	3934.759599	676.725075	14.67
4120	4620.544366	3942.879626	677.664740	14.67
4130	4629.888783	3950.941571	678.947212	14.66

4140	4639.715381	3959.143911	680.571470	14.67
4150	4648.564887	3967.341038	681.223848	14.65
4160	4658.394950	3975.680479	682.714471	14.66
4170	4668.055178	3983.889588	684.165590	14.66
4180	4677.930219	3992.139134	685.791085	14.66
4190	4687.072178	4000.362681	686.709497	14.65
4200	4697.354147	4008.629549	688.724598	14.66
4210	4706.468571	4016.930679	689.537892	14.65
4220	4715.952622	4025.273127	690.679494	14.65
4230	4725.487197	4033.662138	691.825059	14.64
4240	4734.527818	4041.955060	692.572757	14.63
4250	4744.421811	4050.407942	694.013869	14.63
4260	4753.918836	4058.690005	695.228831	14.62
4270	4763.158189	4067.071562	696.086627	14.61
4280	4772.644939	4075.583112	697.061827	14.61
4290	4782.356874	4084.010664	698.346210	14.60
4300	4791.291171	4092.470885	698.820286	14.59
4310	4800.671002	4100.924059	699.746944	14.58
4320	4810.531881	4109.402201	701.129680	14.57
4330	4819.754628	4117.896783	701.857845	14.56
4340	4829.386310	4126.367369	703.018941	14.56
4350	4838.660484	4134.927584	703.732900	14.54
4360	4848.107405	4143.478635	704.628770	14.53
4370	4857.815380	4152.005622	705.809758	14.53
4380	4866.954021	4160.629821	706.324201	14.51
4390	4876.743142	4169.244147	707.498995	14.51
4400	4886.270479	4177.825610	708.444868	14.50
4410	4895.963740	4186.461369	709.502370	14.49
4420	4905.142489	4195.172772	709.969717	14.47
4430	4914.792646	4203.784506	711.008140	14.47
4440	4924.317098	4212.502090	711.815008	14.46
4450	4933.068126	4221.167933	711.900193	14.43
4460	4942.976896	4229.911863	713.065034	14.43
4470	4951.878798	4238.727586	713.151212	14.40
4480	4961.688463	4247.399075	714.289388	14.40
4490	4971.339246	4256.215415	715.123830	14.38
4500	4980.118763	4264.964357	715.154406	14.36
4510	4990.345480	4273.742013	716.603467	14.36
4520	4999.329681	4282.488511	716.841170	14.34
4530	5008.626200	4291.311935	717.314265	14.32
4540	5018.084914	4300.197375	717.887538	14.31
4550	5027.221564	4309.028277	718.193287	14.29
4560	5036.841667	4317.979730	718.861937	14.27
4570	5045.941676	4326.859729	719.081947	14.25
4580	5055.192672	4335.763848	719.428824	14.23
4590	5064.628973	4344.668025	719.960948	14.22
4600	5074.356062	4353.660051	720.696011	14.20
4610	5083.769849	4362.554424	721.215425	14.19
4620	5092.837760	4371.522243	721.315517	14.16
4630	5102.645949	4380.591721	722.054227	14.15
4640	5111.503125	4389.518629	721.984496	14.12
4650	5121.083490	4398.550206	722.533284	14.11
4660	5131.043899	4407.570606	723.473293	14.10
4670	5139.679126	4416.531820	723.147305	14.07
4680	5149.768260	4425.768940	723.999320	14.06
4690	5158.954824	4434.755036	724.199788	14.04

4700	5167.990828	4443.954462	724.036366	14.01
4710	5177.609630	4452.971506	724.638124	14.00
4720	5186.709473	4462.103843	724.605630	13.97
4730	5196.707324	4471.304706	725.402618	13.96
4740	5205.900596	4480.439530	725.461066	13.94
4750	5214.775253	4489.640211	725.135042	13.91
4760	5224.457463	4498.837569	725.619894	13.89
4770	5233.658115	4507.997707	725.660408	13.87
4780	5242.792985	4517.212664	725.580321	13.84
4790	5252.522038	4526.401223	726.120814	13.82
4800	5261.180476	4535.730455	725.450021	13.79
4810	5270.886228	4544.950156	725.936072	13.77
4820	5280.315446	4554.296930	726.018516	13.75
4830	5289.353245	4563.615917	725.737329	13.72
4840	5299.274679	4572.877085	726.397594	13.71
4850	5308.072790	4582.310796	725.761994	13.67
4860	5317.139735	4591.524603	725.615132	13.65
4870	5326.867764	4600.938601	725.929163	13.63
4880	5335.947926	4610.262581	725.685345	13.60
4890	5345.528297	4619.708143	725.820154	13.58
4900	5354.773480	4629.072145	725.701335	13.55
4910	5363.717280	4638.559005	725.158275	13.52
4920	5373.219973	4647.932440	725.287533	13.50
4930	5382.809993	4657.366756	725.443238	13.48
4940	5391.505966	4666.897549	724.608416	13.44
4950	5401.194496	4676.431615	724.762881	13.42
4960	5410.101872	4685.920732	724.181140	13.39
4970	5419.647914	4695.450053	724.197861	13.36
4980	5429.307069	4704.824767	724.482302	13.34
4990	5438.486078	4714.459892	724.026186	13.31
5000	5447.861998	4724.019128	723.842871	13.29
5010	5457.254336	4733.597314	723.657022	13.26
5020	5466.418306	4743.226502	723.191804	13.23
5030	5475.749437	4752.792628	722.956809	13.20
5040	5485.379530	4762.423472	722.956059	13.18
5050	5494.038481	4772.097757	721.940724	13.14
5060	5503.522536	4781.655387	721.867149	13.12
5070	5513.109958	4791.414029	721.695929	13.09
5080	5522.249631	4801.086613	721.163018	13.06
5090	5531.929613	4810.747989	721.181623	13.04
5100	5540.694583	4820.489497	720.205086	13.00
5110	5549.834823	4830.187897	719.646926	12.97
5120	5559.248179	4839.923382	719.324796	12.94
5130	5568.798580	4849.683542	719.115038	12.91
5140	5577.672277	4859.464921	718.207356	12.88
5150	5587.160081	4869.177030	717.983051	12.85
5160	5596.227737	4878.996170	717.231567	12.82
5170	5605.644654	4888.825970	716.818684	12.79
5180	5614.492288	4898.659804	715.832484	12.75
5190	5623.777120	4908.539683	715.237437	12.72
5200	5632.847100	4918.363053	714.484047	12.68
5210	5642.463399	4928.265774	714.197625	12.66
5220	5651.883418	4938.187116	713.696303	12.63
5230	5660.630423	4948.048568	712.581855	12.59
5240	5669.978564	4957.951295	712.027269	12.56
5250	5679.113904	4967.911190	711.202714	12.52

5260	5688.363844	4977.838339	710.525505	12.49
5270	5697.902080	4987.795065	710.107015	12.46
5280	5706.596076	4997.735501	708.860575	12.42
5290	5716.333560	5007.685717	708.647844	12.40
5300	5725.671141	5017.732362	707.938779	12.36
5310	5734.289093	5027.722195	706.566897	12.32
5320	5743.891994	5037.791652	706.100342	12.29
5330	5752.725353	5047.771014	704.954340	12.25
5340	5762.451907	5057.884246	704.567661	12.23
5350	5771.717743	5067.951339	703.766404	12.19
5360	5780.378597	5078.074439	702.304159	12.15
5370	5789.954351	5088.165039	701.789312	12.12
5380	5798.939664	5098.262285	700.677379	12.08
5390	5807.786383	5108.354504	699.431879	12.04
5400	5817.687142	5118.562351	699.124791	12.02
5410	5826.437166	5128.658213	697.778953	11.98
5420	5835.952598	5138.853780	697.098817	11.94
5430	5845.043541	5149.061047	695.982493	11.91
5440	5854.191540	5159.217165	694.974375	11.87
5450	5863.094039	5169.481977	693.612061	11.83
5460	5872.402795	5179.678684	692.724111	11.80
5470	5881.825024	5189.921629	691.903395	11.76
5480	5890.842689	5200.220462	690.622227	11.72
5490	5900.026131	5210.426277	689.599854	11.69
5500	5909.432316	5220.713821	688.718495	11.65
5510	5917.975418	5231.009462	686.965956	11.61
5520	5927.479356	5241.310921	686.168435	11.58
5530	5936.465147	5251.627292	684.837855	11.54
5540	5946.143332	5262.026604	684.116728	11.51
5550	5955.342138	5272.308308	683.033830	11.47
5560	5964.146533	5282.743154	681.403380	11.42
5570	5973.620902	5293.063450	680.557452	11.39
5580	5982.914977	5303.456852	679.458125	11.36
5590	5991.472985	5313.893170	677.579816	11.31
5600	6001.495519	5324.310734	677.184785	11.28
5610	6010.190820	5334.755714	675.435106	11.24
5620	6019.650286	5345.233898	674.416388	11.20
5630	6029.083879	5355.562335	673.521544	11.17
5640	6037.621815	5366.102461	671.519354	11.12
5650	6046.796932	5376.598595	670.198337	11.08
5660	6056.260886	5387.105627	669.155259	11.05
5670	6065.104447	5397.584123	667.520323	11.01
5680	6074.835752	5408.080697	666.755054	10.98
5690	6083.644017	5418.598277	665.045740	10.93
5700	6093.192335	5429.196676	663.995659	10.90
5710	6101.542636	5439.751244	661.791392	10.85
5720	6110.830996	5450.307419	660.523577	10.81
5730	6120.281535	5460.988344	659.293191	10.77
5740	6129.268752	5471.549313	657.719439	10.73
5750	6138.445601	5482.197303	656.248298	10.69
5760	6147.070023	5492.810709	654.259314	10.64
5770	6156.737464	5503.472244	653.265221	10.61
5780	6166.200513	5514.150050	652.050463	10.57
5790	6174.747542	5524.748822	649.998720	10.53
5800	6184.336317	5535.407038	648.929279	10.49
5810	6193.133431	5546.225211	646.908220	10.45

5820	6201.832570	5556.905551	644.927019	10.40
5830	6211.207817	5567.605793	643.602023	10.36
5840	6220.276866	5578.222172	642.054694	10.32
5850	6229.761584	5589.127931	640.633653	10.28
5860	6238.507790	5599.888197	638.619593	10.24
5870	6247.765900	5610.652071	637.113829	10.20
5880	6256.737988	5621.458262	635.279726	10.15
5890	6265.972125	5632.148520	633.823605	10.12
5900	6275.133982	5643.005797	632.128185	10.07
5910	6284.054280	5653.823007	630.231274	10.03
5920	6293.297336	5664.691254	628.606081	9.99
5930	6302.230260	5675.512951	626.717309	9.94
5940	6311.201066	5686.390032	624.811034	9.90
5950	6320.918880	5697.279091	623.639789	9.87
5960	6329.665095	5708.201314	621.463782	9.82
5970	6339.041957	5719.058678	619.983279	9.78
5980	6347.512775	5730.020906	617.491869	9.73
5990	6356.984107	5740.960865	616.023241	9.69
6000	6365.852321	5751.863113	613.989208	9.65
6010	6375.032405	5762.813968	612.218437	9.60
6020	6384.459219	5773.804693	610.654526	9.56
6030	6392.718575	5784.738132	607.980443	9.51
6040	6402.175291	5795.688853	606.486438	9.47
6050	6411.384469	5806.772730	604.611739	9.43
6060	6420.182579	5817.759470	602.423109	9.38
6070	6429.468849	5828.798580	600.670268	9.34
6080	6438.740481	5839.861703	598.878778	9.30
6090	6447.474618	5850.911529	596.563089	9.25
6100	6456.243758	5861.987170	594.256589	9.20
6110	6465.743874	5873.106914	592.636960	9.17
6120	6475.021493	5884.083680	590.937814	9.13
6130	6483.450190	5895.209208	588.240982	9.07
6140	6492.813487	5906.337797	586.475690	9.03
6150	6501.665184	5917.411952	584.253232	8.99
6160	6510.892054	5928.616910	582.275144	8.94
6170	6520.320493	5939.771300	580.549193	8.90
6180	6529.103610	5950.923856	578.179754	8.86
6190	6538.142600	5962.139592	576.003009	8.81
6200	6546.926187	5973.280146	573.646041	8.76
6210	6556.290164	5984.444472	571.845692	8.72
6220	6565.225897	5995.733767	569.492130	8.67
6230	6574.230085	6006.866802	567.363282	8.63
6240	6583.647990	6018.171533	565.476457	8.59
6250	6592.144011	6029.363382	562.780629	8.54
6260	6601.347458	6040.661156	560.686302	8.49
6270	6610.094416	6051.930413	558.164003	8.44
6280	6619.598204	6063.253694	556.344509	8.40
6290	6628.061837	6074.472985	553.588852	8.35
6300	6637.625364	6085.789782	551.835581	8.31
6310	6646.611796	6097.077102	549.534693	8.27
6320	6655.334573	6108.393936	546.940638	8.22
6330	6664.872917	6119.737248	545.135669	8.18
6340	6673.539418	6131.068052	542.471366	8.13
6350	6682.565640	6142.489797	540.075844	8.08
6360	6691.704920	6153.830972	537.873948	8.04
6370	6700.871368	6165.201762	535.669606	7.99

6380	6709.726992	6176.571870	533.155122	7.95
6390	6718.186520	6188.061658	530.124861	7.89
6400	6727.861353	6199.470596	528.390757	7.85
6410	6736.409605	6210.895998	525.513607	7.80
6420	6745.670850	6222.302958	523.367892	7.76
6430	6754.800870	6233.820747	520.980123	7.71
6440	6763.550805	6245.217454	518.333351	7.66
6450	6772.495693	6256.707092	515.788601	7.62
6460	6781.893988	6268.166697	513.727290	7.57
6470	6791.028915	6279.701816	511.327100	7.53
6480	6799.273322	6291.195506	508.077817	7.47
6490	6808.690302	6302.774945	505.915356	7.43
6500	6817.957478	6314.269238	503.688241	7.39
6510	6826.694559	6325.783449	500.911110	7.34
6520	6835.801628	6337.333175	498.468453	7.29
6530	6844.542069	6348.964267	495.577803	7.24
6540	6853.366485	6360.569426	492.797059	7.19
6550	6862.251552	6372.118060	490.133492	7.14
6560	6871.789074	6383.715918	488.073156	7.10
6570	6880.377942	6395.289769	485.088173	7.05
6580	6889.289681	6406.866316	482.423365	7.00
6590	6898.502343	6418.560691	479.941652	6.96
6600	6907.140033	6430.112408	477.027625	6.91
6610	6916.245064	6441.770042	474.475022	6.86
6620	6924.460428	6453.569757	470.890671	6.80
6630	6934.184536	6465.122565	469.061971	6.76
6640	6942.947799	6476.851087	466.096711	6.71
6650	6951.731757	6488.508207	463.223550	6.66
6660	6960.901480	6500.231574	460.669906	6.62
6670	6969.932238	6512.009826	457.922412	6.57
6680	6978.842393	6523.683242	455.159151	6.52
6690	6987.254126	6535.400237	451.853890	6.47
6700	6996.886976	6547.151919	449.735057	6.43
6710	7005.100565	6558.942274	446.158290	6.37
6720	7014.629880	6570.665767	443.964113	6.33
6730	7024.042832	6582.506666	441.536166	6.29
6740	7032.326789	6594.255874	438.070915	6.23
6750	7041.380467	6606.065678	435.314789	6.18
6760	7050.297213	6617.887626	432.409587	6.13
6770	7059.889597	6629.734566	430.155031	6.09
6780	7068.407665	6641.592591	426.815073	6.04
6790	7077.065912	6653.365253	423.700659	5.99
6800	7086.306731	6665.253497	421.053234	5.94
6810	7095.085720	6677.085633	418.000088	5.89
6820	7104.510386	6688.925280	415.585106	5.85
6830	7112.700332	6700.874662	411.825670	5.79
6840	7122.205852	6712.738307	409.467545	5.75
6850	7130.710717	6724.588163	406.122554	5.70
6860	7139.857139	6736.540469	403.316670	5.65
6870	7149.102831	6748.554391	400.548440	5.60
6880	7157.560009	6760.451560	397.108449	5.55
6890	7166.532447	6772.362691	394.169756	5.50
6900	7175.547479	6784.276767	391.270711	5.45
6910	7184.720860	6796.298411	388.422449	5.41
6920	7193.221050	6808.191899	385.029151	5.35
6930	7202.481415	6820.249327	382.232088	5.31

6940	7211.335323	6832.271267	379.064057	5.26
6950	7220.432148	6844.163867	376.268281	5.21
6960	7228.963985	6856.261053	372.702932	5.16
6970	7237.923816	6868.207973	369.715842	5.11
6980	7247.172743	6880.278177	366.894566	5.06
6990	7255.894616	6892.273668	363.620948	5.01
7000	7264.748248	6904.416093	360.332155	4.96
7010	7273.701234	6916.453999	357.247235	4.91
7020	7282.817682	6928.560298	354.257384	4.86
7030	7291.290044	6940.650422	350.639622	4.81
7040	7300.763477	6952.652635	348.110842	4.77
7050	7309.497904	6964.778040	344.719864	4.72
7060	7318.351003	6976.886562	341.464440	4.67
7070	7327.534498	6988.984638	338.549860	4.62
7080	7336.016820	7001.190913	334.825907	4.56
7090	7345.413384	7013.282722	332.130663	4.52
7100	7353.913948	7025.404369	328.509579	4.47
7110	7363.102591	7037.617633	325.484958	4.42
7120	7371.987961	7049.732065	322.255896	4.37
7130	7380.464012	7061.941685	318.522327	4.32
7140	7389.584509	7074.136976	315.447533	4.27
7150	7398.751921	7086.265701	312.486220	4.22
7160	7407.782773	7098.517028	309.265745	4.17
7170	7416.269504	7110.817668	305.451836	4.12
7180	7425.306206	7122.912787	302.393419	4.07
7190	7434.041575	7135.207922	298.833653	4.02
7200	7443.171161	7147.419820	295.751341	3.97
7210	7451.287377	7159.613148	291.674229	3.91
7220	7460.742319	7171.908447	288.833872	3.87
7230	7469.742313	7184.148331	285.593982	3.82
7240	7477.990220	7196.407625	281.582595	3.77
7250	7486.980575	7208.724011	278.256564	3.72
7260	7495.620536	7221.005059	274.615477	3.66
7270	7505.007628	7233.342766	271.664862	3.62
7280	7513.256479	7245.706248	267.550230	3.56
7290	7522.235956	7257.949865	264.286091	3.51
7300	7531.443038	7270.274042	261.168996	3.47
7310	7540.227540	7282.575694	257.651846	3.42
7320	7548.467009	7294.960203	253.506806	3.36
7330	7557.731650	7307.302319	250.429331	3.31
7340	7566.845322	7319.736405	247.108917	3.27
7350	7575.696668	7332.064588	243.632080	3.22
7360	7584.486960	7344.478662	240.008297	3.16
7370	7593.098314	7356.786293	236.312021	3.11
7380	7602.405163	7369.236372	233.168792	3.07
7390	7610.854490	7381.604096	229.250394	3.01
7400	7619.749442	7393.990183	225.759259	2.96
7410	7628.742026	7406.469825	222.272202	2.91
7420	7637.940366	7418.843862	219.096504	2.87
7430	7646.560906	7431.355124	215.205782	2.81
7440	7655.121495	7443.794503	211.326991	2.76
7450	7664.203009	7456.263835	207.939175	2.71
7460	7673.103534	7468.699081	204.404453	2.66
7470	7681.945482	7481.140032	200.805450	2.61
7480	7690.429651	7493.646330	196.783320	2.56
7490	7700.036048	7506.161646	193.874402	2.52

7500	7708.220973	7518.527363	189.693609	2.46
7510	7717.430626	7531.189415	186.241211	2.41
7520	7725.881075	7543.565297	182.315778	2.36
7530	7734.799052	7556.269342	178.529710	2.31
7540	7744.168742	7568.693896	175.474846	2.27
7550	7752.412059	7581.279029	171.133030	2.21
7560	7761.852016	7593.791109	168.060907	2.17
7570	7770.261244	7606.331958	163.929285	2.11
7580	7779.147738	7618.870389	160.277350	2.06
7590	7787.847577	7631.441140	156.406437	2.01
7600	7796.882751	7644.122599	152.760152	1.96
7610	7805.651244	7656.554763	149.096481	1.91
7620	7814.556090	7669.177150	145.378941	1.86
7630	7822.719034	7681.818094	140.900940	1.80
7640	7832.274463	7694.435923	137.838540	1.76
7650	7840.782349	7707.116744	133.665605	1.70
7660	7849.322974	7719.661759	129.661215	1.65
7670	7857.998248	7732.308610	125.689638	1.60
7680	7867.142338	7745.027138	122.115200	1.55
7690	7876.081256	7757.583300	118.497957	1.50
7700	7884.386136	7770.255642	114.130494	1.45
7710	7893.753776	7782.868536	110.885240	1.40
7720	7902.431940	7795.585068	106.846872	1.35
7730	7911.290475	7808.235225	103.055250	1.30
7740	7919.702077	7820.882670	98.819408	1.25
7750	7928.684235	7833.588382	95.095852	1.20
7760	7937.745181	7846.342302	91.402878	1.15
7770	7946.220788	7859.002836	87.217952	1.10
7780	7954.747609	7871.749586	82.998022	1.04
7790	7964.203601	7884.564769	79.638832	1.00
7800	7972.995332	7897.242848	75.752484	0.95
7810	7981.677159	7909.979947	71.697212	0.90
7820	7990.464730	7922.669112	67.795618	0.85
7830	7998.960516	7935.377033	63.583483	0.79
7840	8008.486686	7948.190957	60.295729	0.75
7850	8016.491200	7960.960058	55.531142	0.69
7860	8025.752453	7973.724363	52.028090	0.65
7870	8034.386241	7986.544656	47.841585	0.60
7880	8043.228212	7999.289509	43.938703	0.55
7890	8051.826701	8012.153909	39.672792	0.49
7900	8060.656751	8024.870392	35.786359	0.44
7910	8069.620474	8037.640756	31.979718	0.40
7920	8078.331291	8050.512394	27.818897	0.34
7930	8087.201333	8063.386017	23.815316	0.29
7940	8095.644650	8076.181743	19.462906	0.24
7950	8104.959581	8088.875232	16.084348	0.20
7960	8113.552643	8101.793526	11.759117	0.14
7970	8122.361423	8114.622438	7.738985	0.10
7980	8130.717736	8127.502491	3.215245	0.04
7990	8139.566538	8140.296251	0.729713	0.01
8000	8148.188406	8153.201460	5.013054	0.06
8010	8157.194765	8166.088070	8.893305	0.11
8020	8165.448818	8178.942449	13.493631	0.17
8030	8174.366181	8191.850318	17.484137	0.21
8040	8183.129990	8204.770152	21.640161	0.26
8050	8192.334650	8217.656187	25.321537	0.31

8060	8200.341952	8230.531064	30.189111	0.37
8070	8209.530686	8243.471617	33.940932	0.41
8080	8218.619573	8256.421546	37.801972	0.46
8090	8227.022032	8269.281376	42.259344	0.51
8100	8235.903078	8282.197154	46.294076	0.56
8110	8244.522473	8295.181436	50.658963	0.61
8120	8253.795262	8308.102918	54.307656	0.66
8130	8262.002122	8321.078877	59.076754	0.72
8140	8271.102989	8334.062732	62.959743	0.76
8150	8279.658220	8346.990284	67.332064	0.81
8160	8288.980436	8359.913385	70.932949	0.86
8170	8297.216961	8372.938046	75.721085	0.91
8180	8306.142916	8385.846691	79.703774	0.96
8190	8314.936715	8398.946409	84.009694	1.01
8200	8323.508788	8411.892969	88.384181	1.06
8210	8332.263071	8424.872455	92.609384	1.11
8220	8341.281555	8437.851934	96.570379	1.16
8230	8349.534979	8450.875086	101.340106	1.21
8240	8358.619426	8463.892911	105.273485	1.26
8250	8367.570203	8476.844736	109.274533	1.31
8260	8375.843575	8489.939882	114.096306	1.36
8270	8384.243265	8502.966950	118.723685	1.42
8280	8393.253383	8516.003502	122.750119	1.46
8290	8402.071249	8529.024084	126.952835	1.51
8300	8410.365241	8542.082126	131.716885	1.57
8310	8419.569109	8555.149134	135.580025	1.61
8320	8428.018573	8568.271029	140.252456	1.66
8330	8437.043924	8581.216382	144.172458	1.71
8340	8445.343743	8594.355089	149.011345	1.76
8350	8454.385594	8607.371091	152.985497	1.81
8360	8462.646579	8620.502917	157.856338	1.87
8370	8472.106757	8633.561556	161.454799	1.91
8380	8480.498098	8646.689596	166.191498	1.96
8390	8489.335844	8659.810813	170.474969	2.01
8400	8497.637356	8672.831839	175.194482	2.06
8410	8506.655630	8685.989119	179.333489	2.11
8420	8515.510306	8699.056809	183.546502	2.16
8430	8524.072960	8712.239917	188.166957	2.21
8440	8532.610707	8725.432838	192.822131	2.26
8450	8541.539983	8738.454509	196.914527	2.31
8460	8550.742265	8751.638491	200.896226	2.35
8470	8558.923461	8764.741261	205.817800	2.40
8480	8567.744540	8777.893413	210.148873	2.45
8490	8576.708935	8791.010022	214.301088	2.50
8500	8585.651463	8804.275098	218.623635	2.55
8510	8594.055749	8817.383232	223.327483	2.60
8520	8602.800931	8830.485905	227.684974	2.65
8530	8610.900549	8843.740325	232.839776	2.70
8540	8619.610075	8856.894225	237.284150	2.75
8550	8628.431748	8870.057881	241.626132	2.80
8560	8637.187564	8883.282211	246.094647	2.85
8570	8645.629539	8896.456049	250.826510	2.90
8580	8654.501099	8909.652647	255.151548	2.95
8590	8663.160875	8922.883112	259.722238	3.00
8600	8671.820092	8936.035891	264.215800	3.05
8610	8680.584051	8949.250841	268.666790	3.10

8620	8689.551335	8962.428290	272.876955	3.14
8630	8697.610486	8975.674680	278.064195	3.20
8640	8707.005237	8988.869644	281.864408	3.24
8650	8715.585780	9002.095544	286.509764	3.29
8660	8724.108198	9015.369493	291.261295	3.34
8670	8732.623378	9028.501961	295.878584	3.39
8680	8741.683326	9041.827683	300.144358	3.43
8690	8750.330087	9055.050416	304.720330	3.48
8700	8758.806686	9068.329252	309.522566	3.53
8710	8767.797893	9081.639052	313.841159	3.58
8720	8776.412827	9094.730515	318.317688	3.63
8730	8785.349006	9108.032211	322.683206	3.67
8740	8793.748088	9121.346373	327.598285	3.73
8750	8802.633958	9134.543012	331.909054	3.77
8760	8810.602647	9147.843578	337.240931	3.83
8770	8819.823257	9161.258037	341.434780	3.87
8780	8828.126900	9174.456236	346.329336	3.92
8790	8836.623445	9187.743635	351.120189	3.97
8800	8845.399046	9201.076137	355.677091	4.02
8810	8854.558251	9214.312319	359.754068	4.06
8820	8862.808651	9227.614661	364.806010	4.12
8830	8871.689921	9240.882458	369.192537	4.16
8840	8880.343782	9254.258421	373.914639	4.21
8850	8888.831197	9267.571503	378.740306	4.26
8860	8897.673494	9280.802489	383.128995	4.31
8870	8906.334723	9294.168546	387.833823	4.35
8880	8914.741189	9307.535006	392.793817	4.41
8890	8923.430369	9320.848461	397.418092	4.45
8900	8932.151532	9334.103695	401.952163	4.50
8910	8941.181572	9347.413465	406.231893	4.54
8920	8949.659175	9360.741943	411.082768	4.59
8930	8958.329613	9374.186357	415.856744	4.64
8940	8966.831853	9387.500587	420.668735	4.69
8950	8976.046813	9400.809768	424.762954	4.73
8960	8984.458470	9414.189294	429.730824	4.78
8970	8993.206523	9427.495533	434.289010	4.83
8980	9001.129405	9440.829886	439.700481	4.88
8990	9010.088963	9454.212427	444.123465	4.93
9000	9018.424476	9467.584014	449.159538	4.98
9010	9027.231505	9480.924070	453.692565	5.03
9020	9035.749059	9494.357488	458.608429	5.08
9030	9044.618071	9507.663583	463.045512	5.12
9040	9052.755417	9521.066210	468.310793	5.17
9050	9062.010375	9534.456155	472.445780	5.21
9060	9070.419821	9547.881577	477.461756	5.26
9070	9078.885069	9561.156308	482.271239	5.31
9080	9087.619989	9574.615468	486.995479	5.36
9090	9096.611589	9587.986138	491.374549	5.40
9100	9105.074921	9601.333266	496.258346	5.45
9110	9113.658797	9614.785509	501.126712	5.50
9120	9122.504965	9628.145841	505.640876	5.54
9130	9130.768159	9641.594355	510.826196	5.59
9140	9140.231970	9654.955670	514.723701	5.63
9150	9148.555047	9668.358295	519.803248	5.68
9160	9157.200854	9681.840977	524.640123	5.73
9170	9165.448663	9695.231037	529.782374	5.78

9180	9174.799448	9708.640769	533.841321	5.82
9190	9182.811174	9722.077625	539.266451	5.87
9200	9191.463119	9735.505459	544.042340	5.92
9210	9200.006746	9748.983351	548.976604	5.97
9220	9208.520414	9762.432456	553.912042	6.02
9230	9216.889163	9775.765599	558.876436	6.06
9240	9226.173216	9789.209771	563.036556	6.10
9250	9234.086402	9802.707748	568.621347	6.16
9260	9243.002727	9816.060900	573.058173	6.20
9270	9251.325332	9829.541181	578.215849	6.25
9280	9260.462686	9842.945362	582.482676	6.29
9290	9268.846664	9856.457373	587.610709	6.34
9300	9277.608768	9869.926699	592.317930	6.38
9310	9285.849431	9883.310572	597.461140	6.43
9320	9294.994414	9896.797902	601.803488	6.47
9330	9303.238853	9910.167704	606.928852	6.52
9340	9312.287090	9923.697498	611.410408	6.57
9350	9320.592913	9937.139835	616.546921	6.61
9360	9329.235940	9950.552673	621.316732	6.66
9370	9337.535506	9964.074363	626.538857	6.71
9380	9346.909626	9977.572159	630.662533	6.75
9390	9355.235106	9991.079595	635.844489	6.80
9400	9363.690017	10004.531319	640.841303	6.84
9410	9371.587600	10017.905652	646.318052	6.90
9420	9380.793874	10031.353911	650.560037	6.94
9430	9388.979170	10044.919297	655.940127	6.99
9440	9397.818268	10058.418519	660.600251	7.03
9450	9406.317820	10071.850344	665.532523	7.08
9460	9415.106792	10085.365602	670.258810	7.12
9470	9423.591009	10098.816395	675.225386	7.17
9480	9432.713826	10112.291227	679.577401	7.20
9490	9440.804070	10125.837817	685.033747	7.26
9500	9449.609725	10139.306303	689.696578	7.30
9510	9457.972397	10152.781453	694.809056	7.35
9520	9466.722587	10166.261053	699.538466	7.39
9530	9475.565521	10179.720327	704.154806	7.43
9540	9484.207298	10193.260475	709.053177	7.48
9550	9492.559852	10206.773434	714.213583	7.52
9560	9501.217268	10220.282750	719.065482	7.57
9570	9509.769914	10233.798101	724.028187	7.61
9580	9518.401462	10247.242978	728.841516	7.66
9590	9526.527981	10260.829832	734.301851	7.71
9600	9535.377270	10274.275007	738.897738	7.75
9610	9543.327342	10287.798770	744.471428	7.80
9620	9552.369737	10301.259788	748.890050	7.84
9630	9560.804276	10314.758684	753.954408	7.89
9640	9569.757584	10328.366565	758.608981	7.93
9650	9578.082269	10341.826691	763.744422	7.97
9660	9586.572222	10355.318006	768.745784	8.02
9670	9594.988286	10368.869677	773.881391	8.07
9680	9604.113456	10382.395308	778.281852	8.10
9690	9612.606350	10395.883796	783.277447	8.15
9700	9621.165759	10409.521758	788.355998	8.19
9710	9629.724366	10422.884238	793.159871	8.24
9720	9638.333970	10436.300851	797.966881	8.28
9730	9646.682120	10449.968334	803.286214	8.33

```
9740  9655.670243 10463.400187   807.729944            8.37
9750  9663.881077 10476.948390   813.067314            8.41
9760  9672.494340 10490.489519   817.995179            8.46
9770  9680.762834 10503.963389   823.200556            8.50
9780  9689.424900 10517.472679   828.047778            8.55
9790  9697.964079 10531.033876   833.069798            8.59
9800  9706.802866 10544.563287   837.760422            8.63
9810  9714.689106 10558.075265   843.386159            8.68
9820  9723.737313 10571.619724   847.882411            8.72
9830  9732.006894 10585.132504   853.125610            8.77
9840  9741.076088 10598.724816   857.648729            8.80
9850  9749.482254 10612.198997   862.716743            8.85
9860  9758.307454 10625.695369   867.387914            8.89
9870  9766.238755 10639.279116   873.040361            8.94
9880  9775.088284 10652.801379   877.713096            8.98
9890  9783.722535 10666.298029   882.575494            9.02
9900  9792.246729 10679.884178   887.637449            9.06
9910  9800.831815 10693.391313   892.559498            9.11
9920  9809.662905 10706.940866   897.277961            9.15
9930  9817.665816 10720.411611   902.745795            9.20
9940  9826.946695 10733.948324   907.001629            9.23
9950  9834.670765 10747.463504   912.792739            9.28
9960  9843.426279 10761.071352   917.645074            9.32
9970  9851.623957 10774.622339   922.998382            9.37
9980  9860.360205 10788.136647   927.776442            9.41
9990  9868.714659 10801.582278   932.867618            9.45
10000  9877.782654 10815.109270   937.326616            9.49
--------------------------------------------------------------------------
--------
Mean absolute difference: 563.179879
Mean relative error: 50.50%
Max difference: 2182.705972
```